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**Commodity Prices: Structural Factors, Financial
Markets and Non-linear Dynamics**

Diego Bastourre / Jorge Carrera / Javier Ibarlucia

September 2010

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INVESTIGACIONES ECONÓMICAS



BANCO CENTRAL
DE LA REPÚBLICA ARGENTINA

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EDITOR'S NOTE

After a period of relative indifference, the issue of commodity prices has regained strength in both the macroeconomic research agenda and the debate of economic policy of Latin American countries.

For instance, this is evident when we consider the different explanations regarding the economic performance of the countries of this region, where a prominent place is always given to commodity prices.

This variable belongs to the classical category of exogenous factors in the sense that in its determination there is a very limited role for internal factors. Due to its importance, all the efforts intending to understand the erratic and volatile dynamics of commodity prices are naturally welcomed.

For the Central Bank of Argentina, the relevance of the subject is clearly visible. Proof of this are the regular publication of the Index of Commodity Prices that reflects the evolution of main exports prices of Argentina and the publications of the Economic Research area on the subject. It is worth mentioning that this paper focuses on the study of global determinants of commodity prices rather than on the peculiarities of the subject for the case of Argentina. However, the main conclusions or policy recommendations could certainly be used for this case.

In this new document of the Central Bank Papers Series, the authors propose to return to the agenda of studying global factors that determine commodity price trends. This research proposal was originally considered by Raúl Prebisch and has been growing and gaining momentum over the last sixty years.

In order to incorporate recent aspects related to development of new financial markets and derivative products connected with commodities, the authors suggest using a model with heterogeneous agents. From this, a theoretical prediction emerges to be tested empirically: the existence of a non-linear adjustment pattern of actual prices towards equilibrium ones which are determined by macroeconomic fundamentals.

To this end, a smooth transition autoregressive model, which is suitable for testing the authors' hypothesis, is estimated. The econometric results show the significance of international factors in determining long-run commodity prices such as the international interest rate, the world demand, the real exchange rate of the dollar or the evolution of world stock markets. The existence of a declining secular trend for commodity prices as suggested by Prebisch and Singer at the beginning of the 1950s is also confirmed.

Regarding the non-linear adjustment pattern, it is verified that high misalignments between spot and equilibrium prices tend to correct relatively fast, whereas small gaps tend to persist over time without any correcting force taking place.

Moreover, the paper analyzes the main consequences for commodity prices of two ongoing structural transformations, namely the production of biodiesel and the rapid growth path followed by India and China.

Finally, we would like to stress that with this type of study the Central Bank keeps on expanding the number of technical instruments designed for a better understanding of the domestic and international context of our economy, resulting in more and better judgment elements to make decisions about monetary and financial policy.

A handwritten signature in black ink, appearing to read 'Jorge Carrera', with a long horizontal stroke extending to the right.

Jorge Carrera
Head of Economic Research

BCRA Paper Series | 6

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Commodity Prices: Structural Factors, Financial Markets and Non-linear Dynamics

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Commodity Prices: Structural Factors, Financial Markets and Non-linear Dynamics*

Diego Bastourre (Central Bank of Argentina) / Jorge Carrera (Central Bank of Argentina) /
Javier Ibarlucia (Central Bank of Argentina)

Executive Summary

The volatile movements in commodity prices during the last years have raised new questions to be discussed in both policy and academic spheres. Up to the financial slump of the second quarter of 2008 commodity prices grew fast for several consecutive years in a highly volatile context. The international financial crisis suddenly broke this trend. However, after the abrupt correction during the most traumatic months of the crisis, commodity prices started a recovery, although the rate of growth was slower than in the previous cycle.

Thus, this paper enters into this re-energized debate by developing a theoretical and empirical model that tries to help in understanding both those factors that affect long-run movements in prices and those that influence their short-run dynamics.

In particular, the so-called financialization of commodities phenomenon is analyzed claiming that the impact of this process might be reflected in short-run movements in prices, but not in long-run or equilibrium values.

In technical terms specifically, the main hypothesis is that the process of financialization of commodities could generate a non-linear adjustment pattern of prices towards their long-run equilibrium values. In order to theoretically support this hypothesis, a model of heterogeneous agents that interact in commodity markets is developed. The empirical evaluation is done with a smooth transition vector autoregressive model.

Concerning long-run fundamentals of prices, this study focuses on the macroeconomic determinants of them. Among fundamentals, there is a clear role for world income (proxy variable for global demand). According to the view of several analysts this demand, would be growing at a pace that is not consistent with the expansion of international supply. The value of the US Dollar also plays an important part in the story. The influence of this variable has been stressed in other historical cycles of commodity prices. Additionally, easy monetary conditions in developed countries and the context of excessive liquidity have been pointed out as important elements by several authors. They would tend to add inflationary pressures that are more rapidly transmitted into goods with flexible prices as it is the case of commodities.

In relation to the empirical front, the estimated model highlights the importance of both the macroeconomic fundamentals included in the long-run relationship and the significance of the hypothesis of non-linear price adjustment to the equilibrium value. The results suggest that large misalignments between spot and fundamental prices tend to be corrected relatively fast, while small gaps persist in time since in this case there are not endogenous correcting forces.

* We are especially grateful to Eric Santor for his insightful comments as discussant in the Workshop "Commodity prices: drivers and economic impacts" organized by Bank of England and European Central Bank (London, February 2009). We also want to thank to Jason Henderson of the Federal Reserve Board of Kansas and staff members of the Research Department for their comments during the Conference "Farming, Finance and the Global Marketplace" (Kansas City, June 8-9, 2010). We thank to José Fanelli, Gustavo Cañonero, Federico Traverso, Horacio Aguirre, Augusto Mercadier and Fernando Toledo for helpful comments and discussion. We are indebted with Mariano Sardi and Emilio Blanco for their helpful research assistance. We are responsible for remaining errors.

Finally, this document covers two processes that were originated thirty years ago, but have intensified during the last decade. These processes cannot be omitted when the objective is to use the empirical results of the paper to project future trends in commodity markets. On the one hand, the issue of biofuels as a substitute of more traditional sources of energy is analyzed. On the other hand, the changing pattern of world demand due to structural transformations that are taking place in increasingly important emerging economies such as China and India is considered.

Regarding policy implications for commodity-dependent developing countries, it must be stressed that commodity price misalignments should be carefully monitored: price reversions tend to be abrupt when the gap between actual and fundamental price is higher than 20-25%.

It is also important to note that factors affecting commodity prices (like real international interest rates and the US real exchange rate) are similar to those that influence capital flows. This explains why it is hard for developing countries to cushion terms of trade shocks with external finance. The same fundamentals that worsen terms of trade affect negatively the access to international credit market.

JEL Classification Numbers: C32, D84, Q11.

1. Introduction

During the two decades following the oil shocks of the seventies the issue of commodity prices lost prominence in academic and policy debates. However, in the last eight years the issue has regained its traditional relevance. Nominal records were broken in almost every month from the last part of 2007 and to the first half of 2008 in main commodity markets such as oil, copper, nickel, soybean or rice just to mention a few. Besides, this rising trend in prices was spread out all over commodity varieties.

It is clear now that policymakers and academics did neither forecast the intensity nor the speed of recent commodity price movements. Therefore, the ongoing research agenda includes two key problems: to understand what elements explain price movements; and to review policy responses in the light of this new scenario. In this paper we cover the first issue of the agenda, although we will point out the main policy challenges at the end of the document.

In order to explain the commodity price movements, we find two stories not necessarily well connected, as Krugman (2008) has advocated.

The first story is basically about fundamentals. It says that world income is growing at a pace which is not matched by the supply side of the commodity markets. China and emerging Asia are the main characters here because their living standards are increasing and they have high income commodity demand elasticity due to Engel law (IMF, 2006; Kaplinsky, 2006; OECD-FAO, 2008). In this view fluctuations in the value of the dollar are also stressed. The influence of this variable over commodity prices has been discussed in other historical booms and busts cycles (Ridler and Yandle, 1972; Dornbusch, 1985; or Borensztein and Reinhart, 1994). Moreover, loose monetary conditions and excess of international liquidity are critical elements of this analysis, since they add inflationary pressures which tends to be reflected rapidly and with more intensity in auction markets like commodity ones (Frankel, 2006; Lipsky, 2008). Finally, if the focus is posited on food commodities it will be necessary to remark biofuels as a new determinant of prices (UNCTAD, 2006; IMF, 2007).

The second story points out speculation as the main driving force of recent commodity ups and downs. It stresses the relevance of the so-called “financialization of commodities”, a process according to which a number of non-conventional actors such as investment banks, hedge-funds or pension funds have been investing in commodity-linked instruments.

Of course it could be asserted, as we do in this paper, that both stories overlap and a complete picture needs to take all these pieces together in a coherent way. We argue that the impact of financialization and speculative activity is reflected on short run price dynamics, but not in the long term equilibrium. Specifically, we propose that financialization generates a non-linear adjustment pattern of commodity prices to its fundamental value.

With this hypothesis in mind we develop, in the theoretical front, a simple heterogeneous agent model in commodity markets that include chartists, fundamentalists and portfolio managers.¹ An outcome of this framework is that price adjustment to equilibrium is reached in a non-linear way, being more intense as long as the past gap between the spot price and the equilibrium price increases.

¹ As we will see in the next chapters, chartists are agents that base their decisions on technical analysis, while fundamentalists form their expectations comparing the evolution of current prices with “equilibrium” prices.

Concerning commodity price equilibrium, we assert that it depends on determinants highlighted by previous literature: world demand, real exchange rate of the United States, real international interest rates and the Prebisch-Singer secular declining trend.

A novel characteristic of this paper is the use of an empirical methodology that allows distinguishing permanent from transitory movements in prices once the long run equilibrium is estimated. Thus, in modeling short run dynamics we make use of a smooth transition autoregressive model (STAR) which is suitable for testing the hypothesis derived from our heterogeneous agent based model in commodity markets.

The intuitive idea of the STAR model is that the discrepancy between current and fundamental prices plays a double role. On the one hand, it is the force that drives price changes in the required direction to close the existing gap as in any traditional error correction model. On the other hand, the misalignment acts also as a transition variable, governing the state of the model. The larger the misalignment is, the faster the speed of convergence will be.

The organization of the paper is as follows. In the next chapter the track of commodity markets is described from a long run perspective. In the first part of this section, a description of the main stylized facts regarding commodity price trends and volatility is presented. After that, we review theoretical literature and empirical estimates regarding long run commodity price determinants. This chapter ends with an estimation of the long run relationship between a commodity price index -that includes agricultural goods and metals- and its fundamentals.

The issue of financialization is discussed in Chapter 3. We conclude from the analysis therein that the evidence does not contradict our hypothesis that financialization is important in influencing short run price dynamics rather than equilibrium levels.

Following this, we present in Chapter 4 a stylized model of heterogeneous agents in commodity markets that gives support to our hypothesis. Also the econometric methodology is summarized in this chapter, emphasizing that vector smooth transition autoregressive models is the suitable empirical strategy in order to test the theoretical predictions of our model.

In the next section the econometric analysis is carried out. We find evidence supporting the hypothesis of non linear adjustment of commodity prices towards their long run value. Hence, the results are in line with the idea that heterogeneous expectations in commodity markets can generate this type of adjustment pattern.

Before finalizing this study with the conclusions and policy lessons, in Chapter 6 we examine two ongoing processes: the production of biofuels and the change in the pattern of international demand of primary goods caused by the structural transformation of China and India. Although to some extent the quantitative effect of these phenomena is captured in the empirical model, there may be doubts about its appropriateness to measure accurately the size of the impact of these processes on the long-term trend of commodity prices. In this sense, the evidence presented in this chapter complements the results of the empirical model.

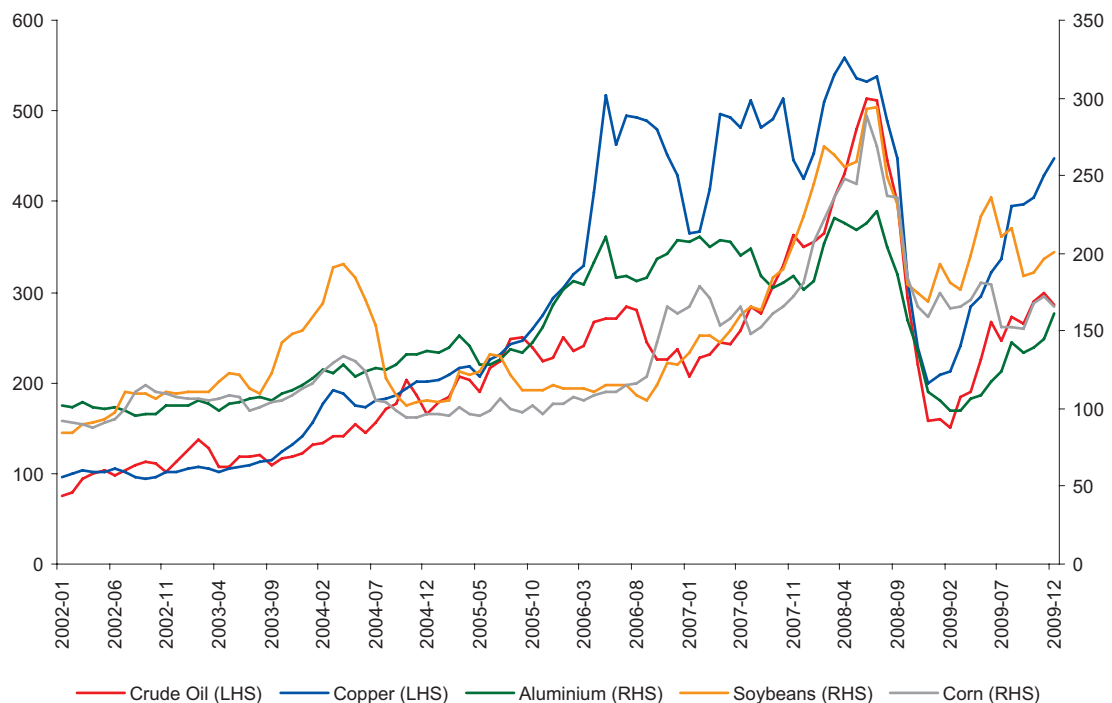
2. Commodity prices in the long-run

2.1. Stylized facts of commodity prices

Up to the intensification of the financial slump during the second part of 2008, commodity prices grew in nominal terms at a strong pace for six consecutive years. However, the crisis revealed once again that commodity price flexibility is remarkable and those record values by June-July of 2008 were cut at least by a 45% at the end of that year and the beginning of 2009. In Figure 1, the evolution of some key commodity prices from 2002 to 2009 is depicted.

Figure 1

Selected commodity price indexes, 2002-1/2009-12 (2002=100)

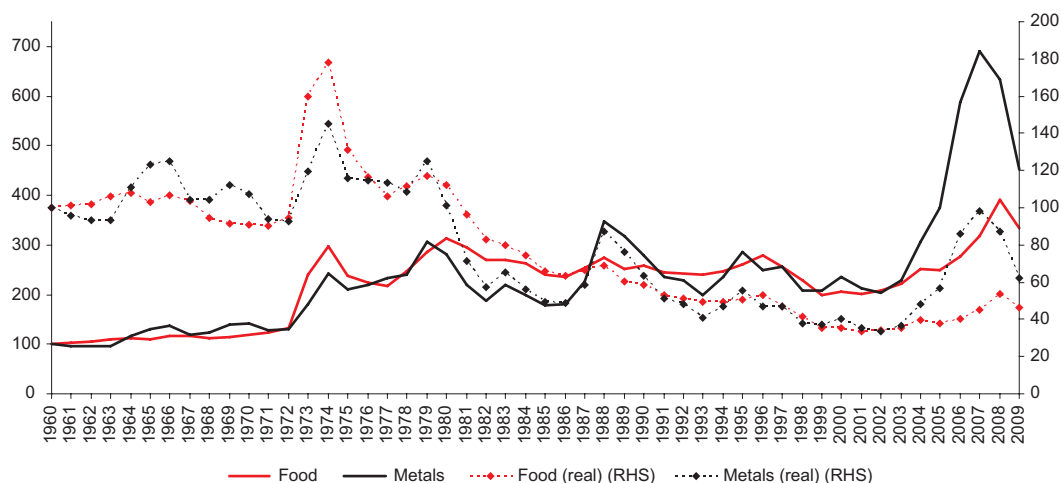


Source: IMF International Financial Statistics.

Contrary to the common belief that commodity prices have reached currently historical high levels, long-run perspective shows stagnant or decaying prices if we incorporate into the analysis the world inflation. In Figures 2 and 3 the long run trends of food, metals and oil in both nominal and real terms are presented. The real series are deflated using US consumer price index (CPI).

Figure 2

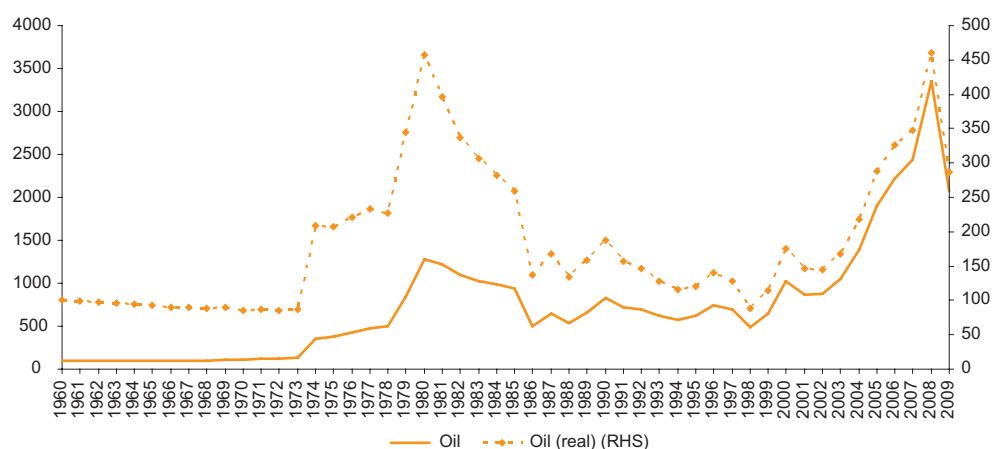
Food and metal price indexes (1960=100)



Source: Author's calculations based on IMF International Financial Statistics data.

Figure 3

Oil price index (1960=100)

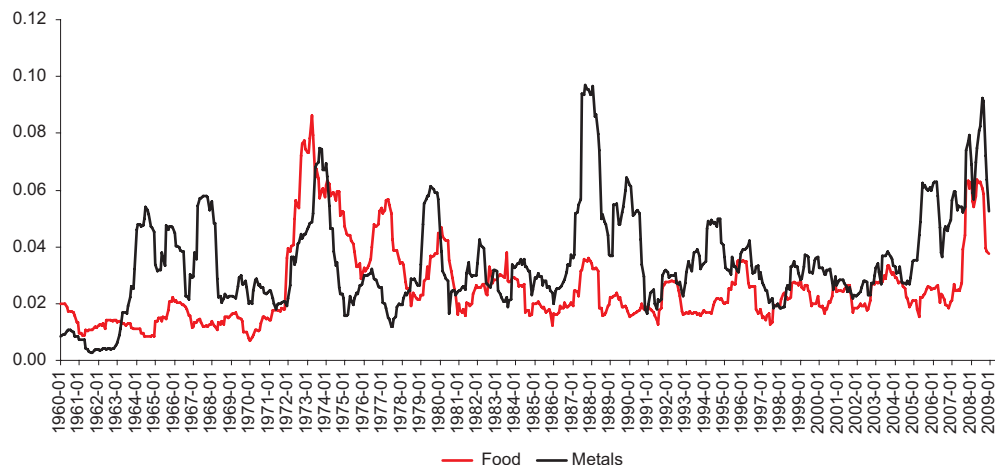


Source: Author's calculations based on IMF International Financial Statistics data.

When food and metals real price indexes are analyzed, we indeed verify between 2002 and 2008 an increase of 170% and 290%, respectively. Nevertheless, when the period 1960-2008 is considered, we observe that even after the last boom, real food prices are far below of their 1960 level (-41.84%) and real metals prices have just recovered the levels exhibited in that year. The oil story is quite different. Real oil price have risen strongly in the last decade, being its level in 2008 five times the figures it presented in 1960.

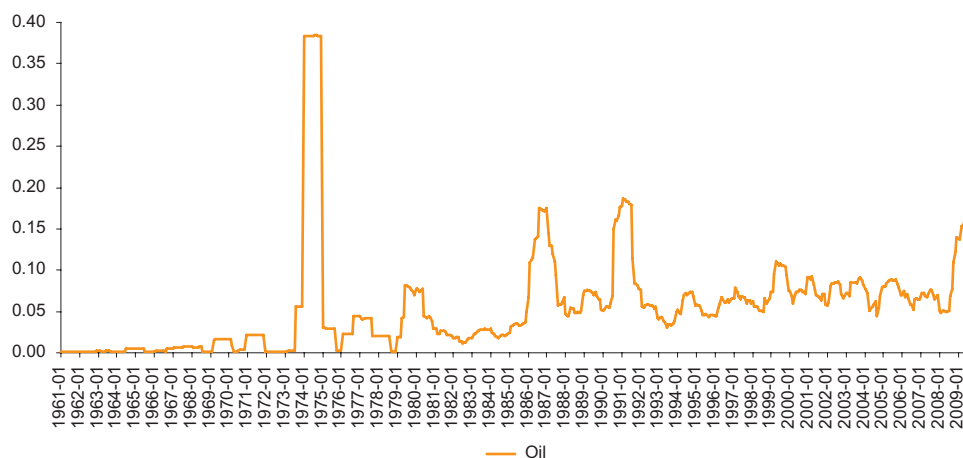
Some authors have stated that, instead of long-run trends, the most remarkable feature of commodity price dynamics is the high short and medium term volatility. According to Deaton (1999) "what commodity prices lack in trend, they make up for in variance". Cashin and McDermott (2002) find that commodity price volatility has increased notably since Bretton Woods breakdown at the beginning of the seventies. Figures 4 and 5

Figure 4
Volatility of food and metal real prices



Source: Author's calculations based on IMF International Financial Statistics data.

Figure 5
Volatility of oil real price



Source: Author's calculations based on IMF International Financial Statistics data.

depict the real price volatility measured by the standard deviation of the monthly real price changes using a rolling window of 12 months.

We found that, in fact, average volatility in post Bretton Woods era has doubled respect to 1960-1972 period in the case of food, while it has risen 40% when metal price index is analyzed. Oil has exhibited a more abrupt price volatility increase. During the first phase the volatility measure averaged 0.0058 whereas in the second one that figure reached 0.069.

After summarizing main stylized facts, the rest of the paper will cover two main issues: i) disentangling the underlying factors that influence commodity prices in the long run –obtaining in this way the “equilibrium” or

“fundamental” price; and ii) studying the mechanisms that generate, strengthen and eventually correct short run deviations of prices with respect to that equilibrium.

2.2. Long run drivers of commodity prices

In one of the most controversial thesis in the international economics field over the past century, Prebisch (1950) and Singer (1950) claimed that, contrary to the classical view, primary product prices would fall relatively to those of the industry. Prebisch (1950) asserted this tendency would be the outcome of a fundamental asymmetry in the international division of labor. Thus, while countries at the “center” had kept all the gains of its productivity increases, “the periphery” had conceded part of the benefits of its own technological progress through lower export good prices.

The influence of this hypothesis over empirical research on commodity prices has been substantial and explains why the primary way of studying these prices has been through univariate methods such as unit root tests or structural breaks tests.²

A different approach for studying commodity prices starts asking which macroeconomic factors play a key role in determining their values.

In this line, the pioneering model of Ridler and Yandle (1972) uses comparative static analysis in a single-good model to demonstrate that an increase in the real value of the dollar (i.e. a real exchange rate appreciation) should result in a fall of dollar commodity prices. Dornbusch (1985) constructs a simple supply-demand two country model to highlight this effect. In that paper, the elasticity of commodity prices to the United States real exchange rate (RER) assumes the following expression:

$$\frac{\partial \ln\left(\frac{P}{CPI}\right)}{\partial \ln\left(\frac{CPI}{eCPI^*}\right)} = -\frac{\beta^*}{\left(\frac{\beta\eta}{\eta^*} + \beta^*\right)} \quad (1)$$

Where P is the price of a representative commodity basket; CPI and CPI^* are consumer price indexes in the United States and the rest of the world respectively; e is the multilateral nominal exchange rate (therefore $\frac{CPI}{eCPI^*}$ is the RER); η and η^* are the demand price elasticities; and β and β^* are the market

shares of each country in the world demand. According to this theoretical model, this elasticity should lie between 0 and -1. However, in empirical research it has usually been the case that it overshoots its theoretical value (i.e. is lower than -1).³

² See for instance Grilli and Yang (1988), Cuddington and Urzúa (1989), Bleaney and Greenaway (1993), Cashing and McDermott (2002) or Ocampo and Parra (2003). The evidence that emerges from these papers is that negative growth rates tend to prevail when commodities are compared to industrial products considering the very long run. However, there is not a clear consensus about growth dynamics. While some authors have argued there is a decaying constant trend, other papers have stressed the importance of structural negative shifts that are not fully recovered during the upward phase of commodity prices cycles.

³ Dornbusch (1985), Gilbert (1989), De Gregorio et al. (2005).

World demand is obviously another important driver of commodity prices. There is general consensus that Engel's law is an accurate framework to predict the impact of income on food commodities (Houthakker, 1987; Hamilton, 2001). Hence, income-elasticity of aggregate food demand in each country would fall as long as the transit to development is completed. In the case of metals, it has been argued that exists an inverse U-shape relationship between its use and income level. Thus, the consumption intensity of metals increases up to a point in which GDP per capita reaches approximately 15,000 or 20,000 PPP (purchasing power parity) adjusted USD (IMF, 2006) and then it starts to go down. In empirical models, indexes of world industrial production have usually been employed to measure global demand.

In addition to US real exchange rate and industrial production, a third variable has been suggested as a key determinant of commodity prices, namely the real interest rate.

Explaining the excess of co-movement of commodity prices with respect to fundamentals, Pindyck and Rotemberg (1987) consider that these movements are the result of herd behavior in financial markets since its participants could believe that all commodities tend to move together. The authors claim that, as storable assets, commodities are affected by expectations. Interest rate might affect the harvest or production in a number of commodities changing its future supply and so current prices. It could also affect expectations about future economic activity and then future commodity demands which, again, impacts on spot prices.

Frankel (2006) remarks that rising interest rates are transmitted to commodity prices through three channels: i) by increasing the incentive for extraction (or production) today rather than tomorrow; ii) by decreasing the desire of firms to carry inventories; and iii) by encouraging speculators to shift out of commodity contracts into treasury bills. The three channels of transmission work to reduce spot prices. In fact, the author has argued that nominal records during 2006 in some commodities could be a signal that monetary policy has been too loose.

In our empirical model we will allow all these variables play an explicit role in determining the "equilibrium" or "fundamental" long run price of commodity prices.

2.3. The long-run empirical model

2.3.1. The variables

Our empirical analysis is based on monthly data belonging to the post Bretton Woods era, from 1973-1 to 2008-5.

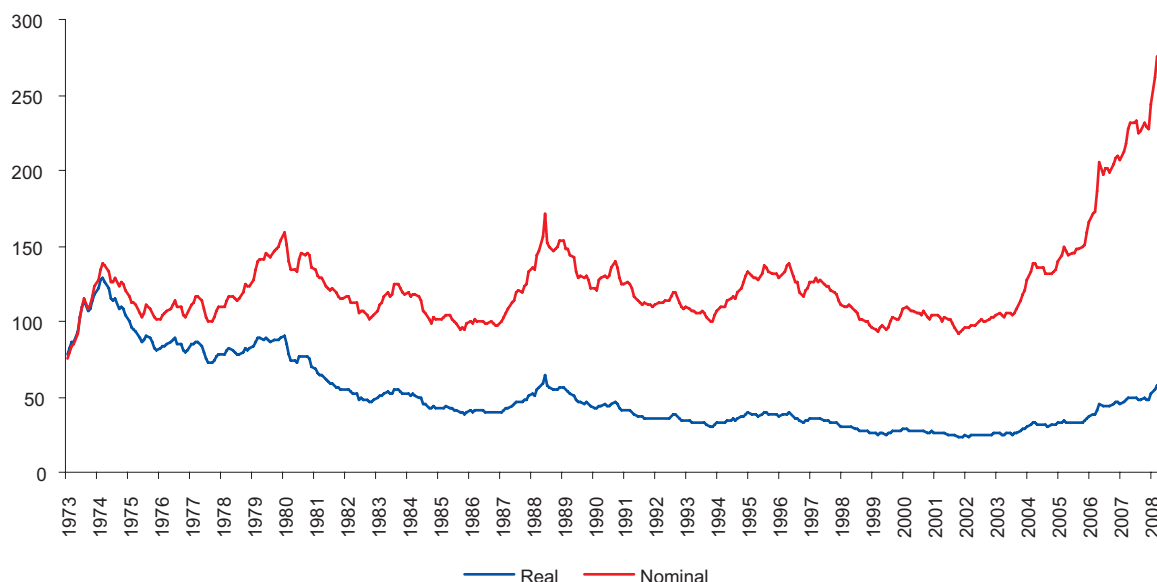
As regards commodity prices to be explained, given that our theoretical framework is general enough and holds for a broad set of assets, we choose to analyze an aggregated index. Particularly, the Food Index and the Metals Index published by International Finance Statistics (IFS-IFM) were averaged out to construct the All-Comm Index. To deflate the series, the US CPI from the same source was used.⁴ Figure 6 shows the evolution of the index in both nominal and real terms.⁵

⁴ It is worth mentioning that oil was not considered because several authors such as Beenstock (1988) or, more recently, Cheung and Morin (2007) have stressed which it has its own dynamics with low connection to other commodities.

⁵ An additional comment related to the issue of commodities viewed as financial assets is needed. The price indexes used in this paper are elaborated with data from spot prices and not from futures or other similar derivative instruments, which are in strict financial terms the real investment vehicles. However, the argument is that spot prices are very good indicators of the financial returns of commodities. In this sense, Gorton and Rouwenhorst (2004) show that the return on a future position and the current price has a correlation close to one.

Figure 6

All-Comm commodity price index (1973=100)



Source: Author's calculations based on IMF International Financial Statistics data.

As it was highlighted in the section 2.1, the outlook for commodities dramatically changes whether nominal or real indexes are considered. Both series have exhibited an important growth since 2002; while the former has increased 166.3% up to 2008 from then, the latter has risen 124.3% in the same time interval. However, the overall performance has been very different: whereas in the first semester of 2008 the nominal index averaged a value 2.6 times higher than the figure of 1973, current real prices are roughly a half of the value observed at the beginning of the sample.

Regarding price fundamentals, the US real exchange rate (*RER*) is considered in the first place. The broad multilateral version published by the Federal Reserve Bank of New York is employed. The variable is defined in such a way that it rises when the dollar is appreciating against the currencies of its main trading partners.

As real international interest rate (*IR*) is utilized the one-year Treasury constant maturity rate from the Board of Governors of the Federal Reserve System. Again, the US CPI is the deflator.

The fundamental series whose measurement presents serious difficulties is the commodity world demand proxy, especially because a series in monthly basis is required. Industrial production index of the developed countries is the alternative usually employed in studies on this topic. However, Asian emerging countries have become crucial players in commodity markets and they should be incorporated into the analysis. Therefore, we construct an Industrial Production Index (*PRO*) which also takes into account the industrial output of South Korea, India, Malaysia and China. All the indexes come from the IFS, except from China whose series is built using the Industrial Value Added provided in an annual-basis by the IMF World Economic Outlook and linearly extrapolated to become it monthly. We use the share of each country in the industrial added value (IVA) as a weighting factor.⁶ The seasonally unadjusted series were corrected applying the X-12 ARIMA method.

⁶ The weights obtained by this criterion were increased by 50% in the case of emerging countries for two reasons. The first one is that emerging economies are underrepresented in the sample. The second reason is that these countries have a greater intensity of commodity consumption per unit

In addition to the price determinants reviewed in the previous section, the Dow Jones Index deflated by the US CPI (*DOW*) is also included. Our aim is to control for the return of alternative assets and to investigate if conditional covariance indicates substitution or complementary effects between these asset classes. Authors like Gorton and Rouwenhorst (2004) or Deutsche Bank (2005) have studied this link and they have found that the non-conditional correlation between returns of commodities and other assets (bonds and equities) is negative and very significant in the long term. In Domanski and Healt (2007) the ratio of net long contracts for non-commercial agents is negatively correlated, though slightly, with stock indexes.

2.3.2. Econometric estimation

The estimation of the long run relationship between commodity prices and their fundamentals requires coping with non stationary series.⁷ As it is well known, if $I(1)$ variables are cointegrated the OLS estimator of the long run coefficients will be consistent. However, it will have a non-normal distribution, and inferences based on t -statistics will be misleading.

Several econometrics techniques have been developed to overcome this problem. We apply the Dynamic OLS (DOLS) estimator developed by Stock and Watson (1993) which adds leads and lags of the changes in the right hand regressors of the standard long run equation. That is,

$$P_t = \beta_1 + \beta_2 X_t + \sum_{j=-p}^p \beta_3 \Delta X_{t-j} + \beta_4 t + \varepsilon_t \quad (2)$$

Where X_t stands for the price determinants and p represents the number of leads and lags considered. DOLS estimator of β_2 results from OLS estimation of equation (2).

If the variables are cointegrated, the DOLS estimator is consistent and efficient in large samples. The methodology deals with potential simultaneity and small sample bias among the regressors by the inclusion of leads and lags. Besides, Monte Carlo experiments show that DOLS performs better, particularly in small samples, compared to alternative estimators of long-run parameters as those proposed by Engle and Granger (1987), Johansen (1988), and Phillips and Hansen (1990).⁸ Finally, standard statistical inference remains valid when heteroskedastic and autocorrelation consistent (HAC) standard errors are employed.

The estimation results of the long run equation coefficients of the commodity prices are shown in Table 1. We use six leads and lags but the estimation was robust to changes in the value of p .

of output. Capturing the specific amount of increase of this coefficient is beyond the scope of this study, but increasing it seems a better alternative than to ignore the two above-mentioned effects.

⁷ The usual unit root tests were implemented confirming all the series are non stationary. Results are available upon request to the authors.

⁸ See Stock and Watson (1993) and Montalvo (1995). It is worth mentioning that DOLS estimator is asymptotically equivalent to the Johansen estimator (Stock and Watson, 1993).

Table 1

DOLS Estimates of long run equation

Variable	OLS	DOLS	
	Coefficient	Coefficient	p-value
Industrial Production Index	3.6124	4.1368	0.0000
US Real Exchange Rate	-0.7270	-0.5994	0.0000
Real International Interest Rate	-1.0907	-1.5452	0.0138
Real Dow Jones Index	-0.1299	-0.1086	0.0163
Time Trend	-0.0097	-0.0111	0.0000
Constant	-4.1659	-6.9366	0.0000

Before analyzing these results it is necessary to perform a cointegration test. If there is no cointegration among variables, it is not possible to evaluate the estimates because of the spurious regression problem.

As it is remarked by Choi et al. (2008), DOLS is employed in many applications but few cointegration tests have been developed for it. These authors propose a Hausman-type test but it does not allow for regressions with time trends like in our model. Therefore, we use the test proposed by Shin (1994) which admits all kind of deterministic components.

An interesting common feature of both tests is that their null hypothesis is the presence of cointegration, in contrast to the no-cointegration null hypothesis of standard ADF-type tests. Shin (1994) and Ogaki and Park (1998) argue that cointegration is the desirable null hypothesis in several applications.

When constant and time trend are present in the long run equation, the Shin cointegration test statistic takes the following expression

$$C_{\tau} = T^{-2} \sum_{t=1}^T S_t^2 / s_{\tau}^2(l) \quad (3)$$

Where $S_t = \sum_{j=1}^t \varepsilon_j$, ε_j are the DOLS residuals from equation (2), and T stands for the sample size. Finally, $s_{\tau}^2(l)$ is the semiparametric consistent estimator of the long run variance of ε_j .

We find a value of 0.07796 for C_{τ} which is lower than the 2.5% critical value (0.088).⁹ Thus, there is evidence of cointegration among commodity prices and their determinants.¹⁰ Then, we can analyze the results showed in Table 1.

In the first place, it is remarkable that all the variables are statistically significant and their signs are in line with the theoretical predictions.

Next, regarding the value of the coefficients, we observe that a real devaluation (fall) of the dollar by 1% implies a 0.60% increase in commodity prices. Thus, this elasticity is lower than one in absolute value as

⁹ Shin (1994) provides the critical values for his cointegration test statistics.

¹⁰ For robustness check, we carried out the standard ADF non-cointegration test based on OLS residuals and we rejected the null hypothesis of unit root residuals.

suggested by the Ridle and Yandle (1972) model. It is also relevant to highlight the high elasticity of commodity prices to industrial production (4.13%).

Stock market performance impacts negatively on the price of commodities. Along the findings by Domanski and Healt (2007), the substitution effect between the two asset classes tends to predominate. As for the coefficient of the real interest rate, it is slightly bigger than -1 in absolute value (-1.09).

Finally, the negative sign of the time trend supports the Prebisch-Singer hypothesis. This result corroborates the intuition suggested by Figure 6, where declining real prices are observed.

In Figure 7, actual and “equilibrium” price series are depicted while implied misalignment (defined as the difference between the former and the latter) is portrayed in Figure 8.

From both figures we emphasize the fact that the magnitude of the misalignments observed during the period 2006-2007 previous to the US subprime crisis are similar to those observed in the previous years to the 80s debt crisis and both Asian and Russian Crisis (roughly 20-25%).

In the fourth section we will see how misalignment plays a dual role: as adjustment variable of short term price deviations from equilibrium ones in one hand, and as a potential determinant of the state or regime of the model, in the other hand.

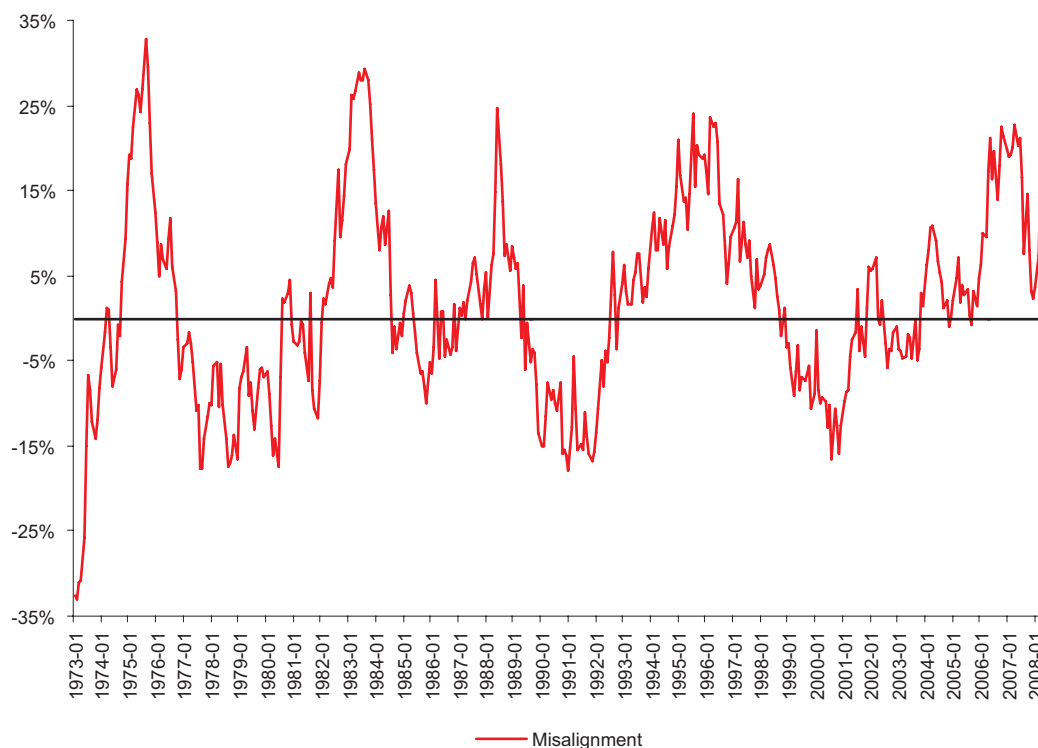
Figure 7

Commodity price index: actual values and equilibrium (logarithmic scale)



Figure 8

Commodity prices: estimated misalignment



3. Financialization of commodity markets

The debate regarding the issue of financialization in commodity markets has been intense and positions are diverse in this point. Some authors have blamed financial markets as the only element responsible for abrupt price ups and downs, while others have neglected their influence on prices. Hence, the key issue among financial market participants, academics and policymakers is trying to establish to what extent the financialization of commodities influence spot price levels and their evolution.

According to their characteristics, the market for each commodity is divided into two types. On the one hand, it is the physical spot market in which the consumers demand these goods to the producer and the spot price is determined. On the other hand, it is the financial derivative market where long and short traders agree on a future settled price. Derivative markets can be further divided in two categories: exchange markets where standardized contracts are traded through a central clearing entity, and over-the-counter markets (OTC) in which tailored contracts are negotiated, usually by means of a market maker.

With the term “financialization” of commodities the literature usually makes reference to two different though partially linked facts. The first fact is that derivative market activity has experienced an impressive growth in the last years. The second issue is the increase in the participation of financial investors in futures markets that occurred simultaneously.

Hence, one of the reasons to believe there is a close connection between price dynamics and speculative activity rests on the idea that we observed, in the 2002-2008 period, a consistent rise in commodity prices in conjunction with a notable increase in turnover on commodity-linked instruments. In fact, trading volume

in these instruments is several times higher than that of the physical production. Just to mention an example, Domanski and Healt (2007) have pointed out that contracts in derivative commodity markets tripled between 2002 and 2005; while in the same period the ratio of financial activity of crude oil and copper to their world productions increased from 3.2 to 3.9 and from 30.5 to 36.1, respectively.

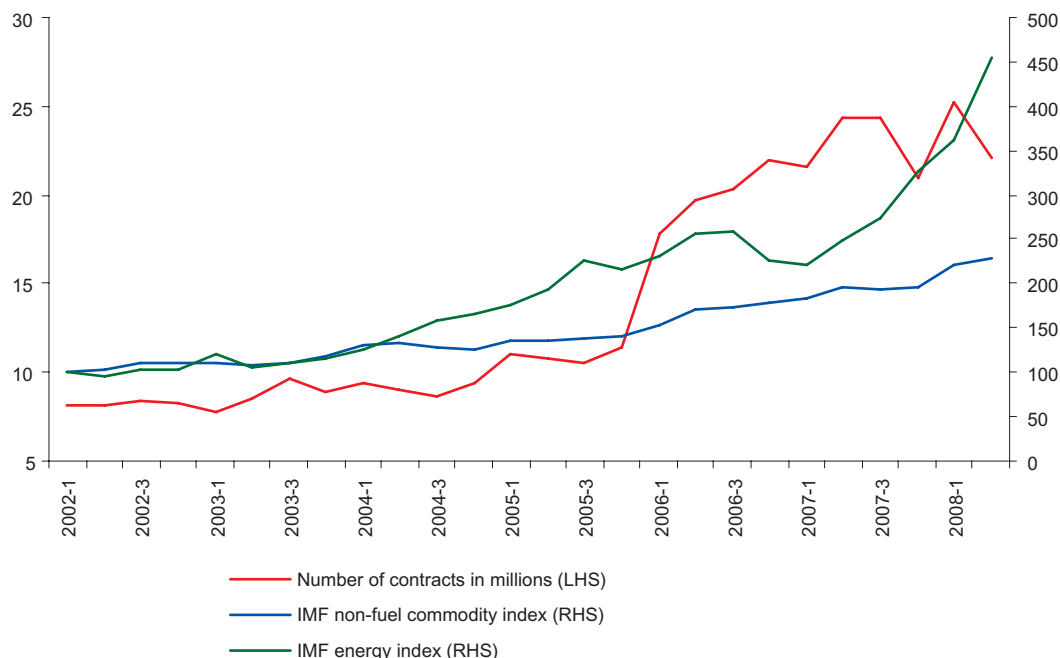
It is possible to illustrate this hypothesis comparing both the evolution of the number of outstanding commodity contracts and the amounts of USD outstanding OTC derivatives in conjunction with nominal commodity price movements. The exercise is presented in Figures 9 and 10 using the energy index and the non-fuel commodity index elaborated by the IMF.

As it was mentioned before, there is evidence of a considerable rise in derivative market activity when the so-called “boom” cycle of commodity prices from the beginning of 2002 to the second quarter of 2008 is observed. During this period, total commodity contracts grew by 170%; while energy and non-energy commodity prices increased by 350% and 120% respectively. Moreover, the evolution of OTC derivatives in nominal USD amounts was even more impressive as Figure 10 reveals.

Thus, in the light of this evidence it results quite natural to associate this significant increase in financial commodity-market deepening with soaring prices.

Figure 9

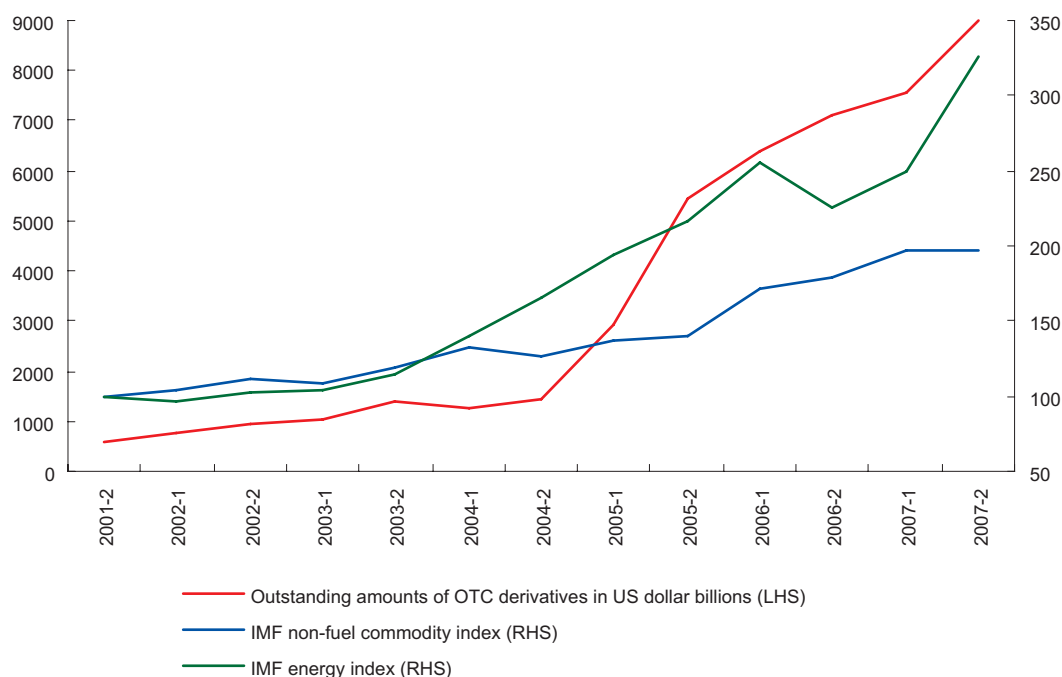
Evolution of derivative commodity contracts and commodity prices, 2002Q1-2008Q2



Source: IMF International Financial Statistics and BIS Quarterly Review.

Figure 10

Evolution of OTC derivatives and commodity prices, 2001S2-2008S1



Source: IMF International Financial Statistics and BIS Quarterly Review.

However, there is an important caveat in this line of reasoning. Specifically, if financialization has actually played a fundamental role in boosting commodity prices, we would expect lower growth rates for those commodities that lack derivative markets. Next figures reproduce the exercises carried out by Deutsche Bank (2008) and Viñals (2008) consisting in calculating price increases in both exchange and non-exchange trade commodities.

In Figure 11, we observe that prices of exchange traded commodities have appreciated by a similar, if not lower, amount to non-exchange traded commodities where financial activity is not possible.

Viñals (2008) has performed a similar exercise considering just the final part of the commodity boom (Figure 12). Again, we cannot establish a clear cut line between exchange and non-exchange traded commodities in terms of price variations. In some sense, this refutes the claim that commodity price increases that took place until the first half of 2008 were driven largely by speculative activity.

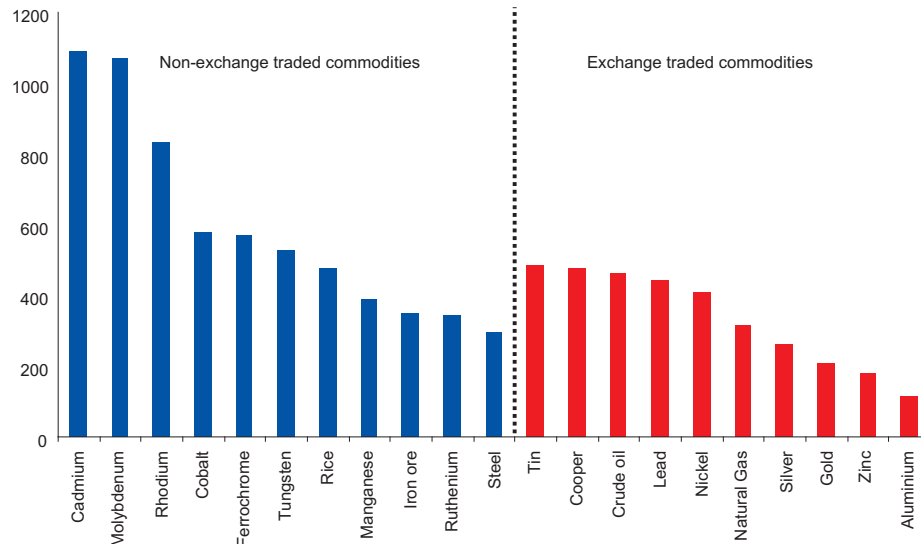
A more careful look to the issue of financialization requires going deep into the microstructure of derivative markets. Among these market participants, there is a first wide division between commercial hedgers and financial investors.

Commercial hedgers are buyer or sellers of the physical commodity who use derivatives to hedge against the risk of price fluctuations. In the end, this type of agent is interested in the evolution of future spot prices of the underlying commodity (IMF, 2006).

Financial participants have different incentives from commercial hedgers. We can distinguish two strategies among them. On the one hand, there are “buy and hold” investors who pursue fully collateralized long-only

Figure 11

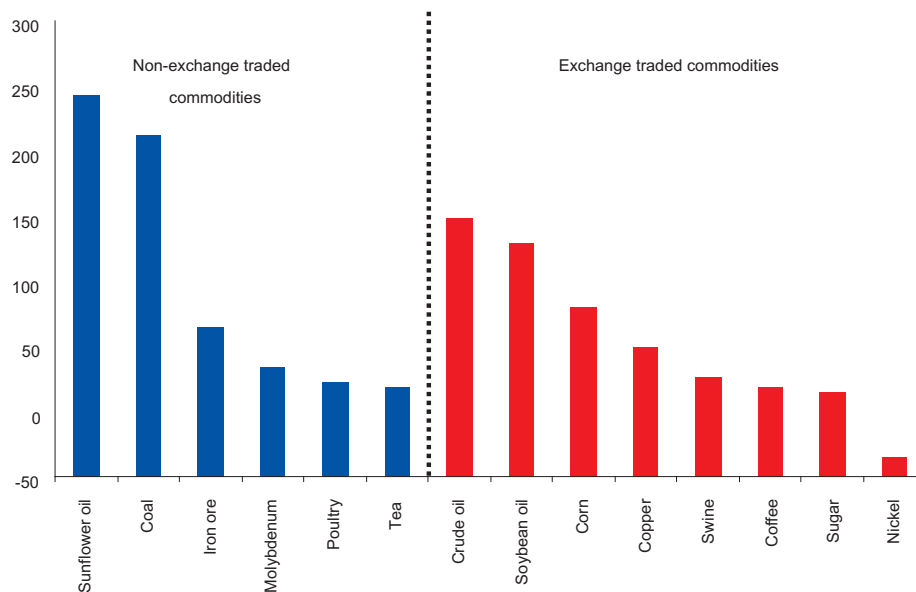
Price changes in selected commodity markets, 2001-2008



Source: Deutsche Bank (2008).

Figure 12.

Price changes in selected commodity markets, 2007-2008



Source: Viñals (2008).

future strategies, i.e., acquiring a long position in futures and investing the same amount in treasury bills as collateral. This strategy is usual among pension and mutual funds and it has historically had excess returns similar to those of equities. Additional interesting properties of commodity futures as a buy and hold strategy is that their returns are negatively correlated with equity and bond returns, and they prove to be a good hedge against unexpected inflation (Gorton and Rouwenhorst, 2004; Erb and Harvey, 2005).

On the other hand, we have a broader group of investors pursuing more complex strategies. For instance, hedge funds are currently playing a very active role. Their operations are characterized by freedom in using wide range of instruments, ability to short sell and high leverage (Stefanini, 2006). Also, retail investors are becoming increasingly important since they could participate in new instruments such as *Exchange Trade Commodity Funds* or *Structured Commodity Notes* (McNee, 2006; Bienkowski, 2007).

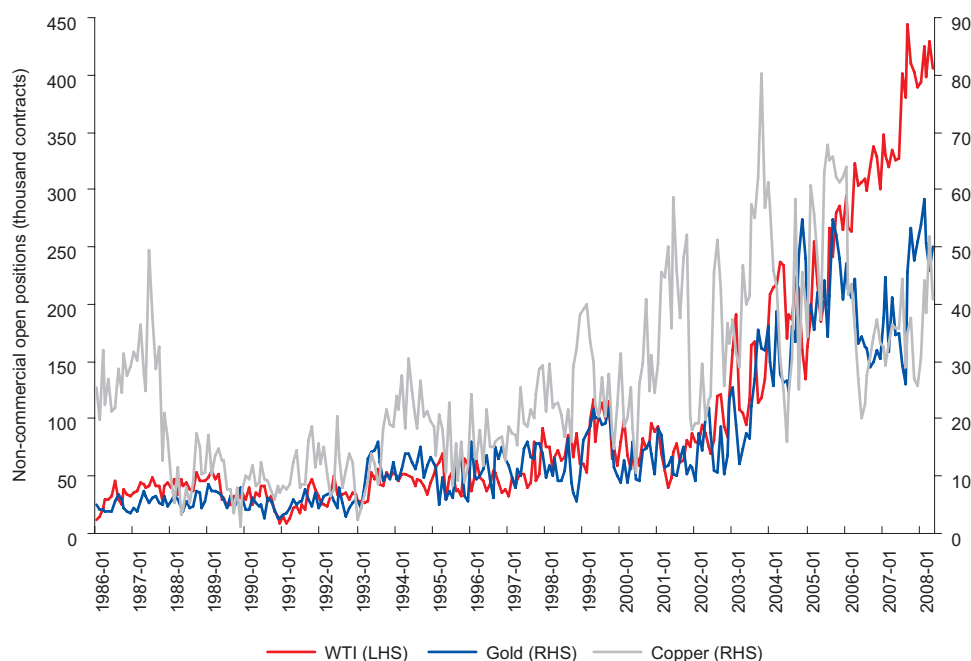
Although it could be argued that the action of financial investors have increased in the last years, this is not entirely a totally new phenomenon. This statement can be verified in Figures 13 and 14, where the number of open positions (long plus short) of financial participants in key future commodity markets since 1986 is depicted.

We observe a strong increase in the number of contracts opened by financial investors in the oil, gold and copper markets since the beginning of 2002; but the activity previous to that date was far from negligible (Figure 13).

The picture is quite different from soft agricultural commodity markets in which we observe a sharp drop in financial activity connected to the Asian crisis of 1998 (Figure 14). After that, the number of open contracts just recovered its previous levels. From this evidence we conclude that financial activity has intensified in the last five years but could not be viewed as a completely new fact.

Figure 13

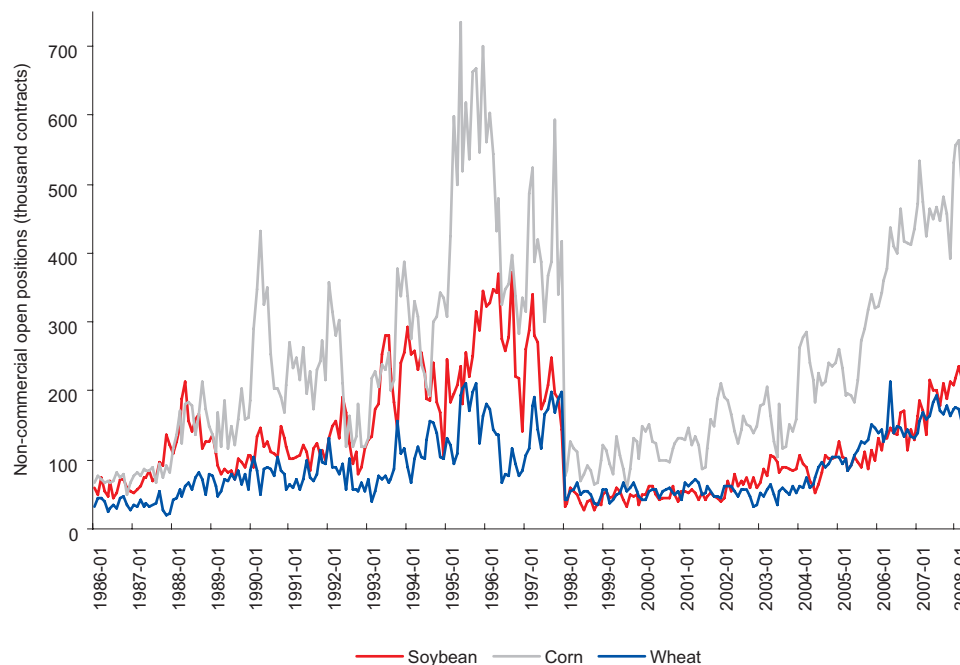
Non-commercial open positions in oil, gold and copper, 1986-1/2008-5



Source: Author's calculations based on Deutsche Bank data.

Figure 14

Non-commercial open positions in soybean, corn and wheat, 1986-1/2008-5



Source: Author's calculations based on Deutsche Bank data.

Besides, the impact of speculative activity on prices could depend more on net positions (long minus short) rather than open positions. In Figure 15 we have drawn these net positions of financial participants in selected derivative markets and the corresponding spot price (expressed as an index).

There are several relevant points to highlight from this figure. Firstly, it seems that the mean position of financial investors throughout time tends to be a long one, but this fact could be partially explained by institutional investors going into long “buy and hold” strategies. There is so a positive “long bias” among non-commercial participants.

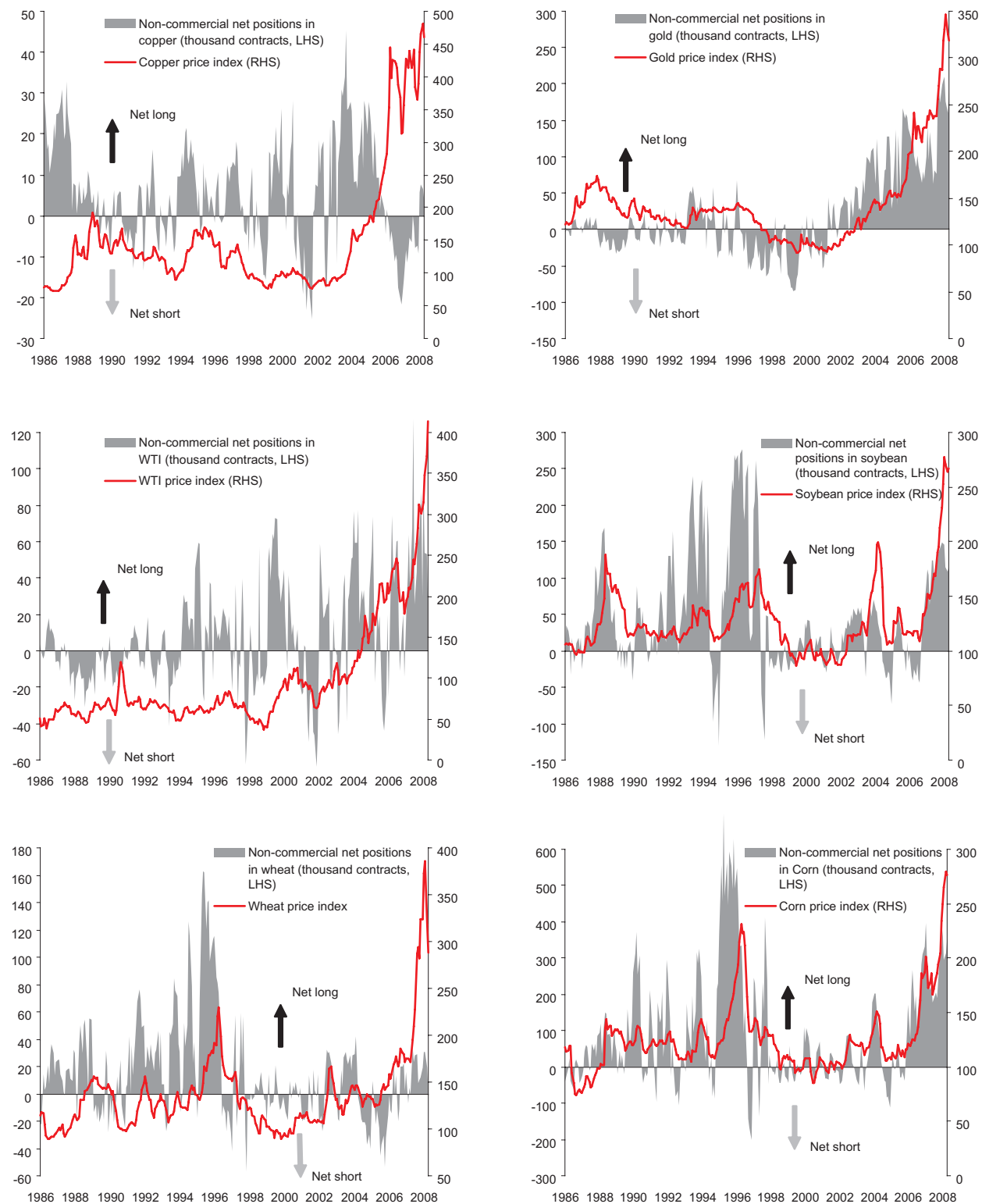
In the second place, we can observe that net financial positions tend to be volatile which means that non-commercial investors could either act as a push or pull factor of influence over prices depending on specific circumstances.

In the third place, it seems there is a positive correlation between the net position and the spot commodity price which indicates that high price levels induce appreciation expectations of financial participants.

Finally, with the notable exception of copper, the last phase of high prices from 2005 went along with short net non-commercial positions. However, it is important to note that apart from the most recent case of copper, there were other phases in the past in which net long positions coexisted with stagnant commodity prices. In wheat for instance, we observed a 200% increase in prices between 2005 and 2008 which coincides with a slight long position during the whole period. Moreover, we could also note from Figure 15 that price drops were not necessarily followed by aggressive net short positions.

Figure 15

Net positions of financial investors in selected commodity markets, 1986-1/2008-5



Source: Author's calculations based on Deutsche Bank and IMF data.

Our reading of the evidence so far is that it is not necessary to have neither derivative markets nor aggressive financial investor participation to observe sharp price fluctuations.

In the financialization debate there is a last remark connected with the issue of causality. We think that a hypothesis that deserves some attention is that higher prices during the past five years could have caused an increased interest in commodity investment, and not the other way around as it is usually stated.

To shed some light about this hypothesis we have calculated in Table 2 the correlation coefficient between net financial positions and the growth rates in various commodity prices using different time windows. We take the net financial positions at the end of the month and compute firstly the contemporaneous correlation with the price variation taking place between this date and the previous month. Then we consider both backward and forward price variation windows up to three-months. In this way, the pairwise correlation between open positions and price changes at $t + 3$ considers the accumulated three month forward price variation. As a complementary exercise we have added the corresponding Granger causality test to establish whether changes in financial positions anticipate spot price changes or if causality runs the other way around.

It is clear that in all the cases there is a strong positive correlation of past price variation at time $(t - 1, t - 2, t - 3)$ that tend to lower in subsequent periods. This fact could indicate that financial speculators take long positions when they observe price increases in the recent past with the expectation of further increases in the future. In the next section, we will refer to this type of behavior as driven by simple “rules of thumb” or “chartist analysis”. The evidence of Granger causality tests, however, does not reveal any clear causality pattern. This means the issue of causality remains open and further research would be needed.

Taking together the pieces of information provided in this section we interpret this preliminary evidence as a signal that financial participants could induce price fluctuations or excess volatility in some markets, but do not have a long-lasting impact on equilibrium prices which are ultimately determined by fundamental supply-demand factors. In the next section, we try to formalize this idea.

Table 2

Net financial positions and commodity prices, correlations and causality, 1986-2008

	COPPER	GOLD	SILVER	WTI	SUGAR	SOYBEAN	CORN	WEATH
T vs $T-3$	0.35	0.64	0.53	0.39	0.09	0.52	0.58	0.34
T vs $T-2$	0.37	0.59	0.50	0.39	0.07	0.51	0.56	0.37
T vs $T-1$	0.38	0.50	0.46	0.40	0.05	0.47	0.50	0.39
T vs T	0.39	0.42	0.44	0.41	0.02	0.40	0.42	0.37
T vs $T+1$	0.22	0.27	0.17	0.16	0.07	0.20	0.26	0.21
T vs $T+2$	0.19	0.29	0.13	0.11	0.15	0.15	0.24	0.20
T vs $T+3$	0.20	0.34	0.14	0.12	0.18	0.12	0.21	0.17
Granger Causality Test								
Price changes do not Granger cause changes in net financial positions	1.77820**	0.67559	0.86989	0.80581	0.30402	0.98956	1.57599*	0.82225
Changes in net financial positions do not Granger cause price changes	0.76235	2.785***	1.05728	1.29875	0.80934	2.09115**	1.21476	1.56979*

Source: Author's calculations based on Deutsche Bank and IMF data.

4. The role of the financialization in the short-run dynamics of commodity prices

In the first part of this chapter we develop a simple model to stress a potential mechanism in which financialization modifies the price adjustment pattern toward the equilibrium. Following this, the econometric methodology which allows evaluating if actual commodity data fits with theoretical model predictions is presented.

4.1. A model of heterogeneous agents in commodity markets

The model assumes that commodity price change of the next period is determined by the interaction of three different agents¹¹ called fundamentalists (F), chartists (C) and portfolio managers (PM) in accordance to the following expression:

$$\Delta P_{t+1} = a_1 E(\Delta P_{t+1}^C) + a_2 E(\Delta P_{t+1}^F) + a_3 E(\Delta P_{t+1}^{PM}) \quad (4)$$

Where $E(\Delta P_{t+1}^C)$, $E(\Delta P_{t+1}^F)$ and $E(\Delta P_{t+1}^{PM})$ are the price change expectations of each agent and a_1, a_2, a_3 are fixed weights that measure the relative importance of each group.

The expectations of the fundamentalists are based on the notion of commodity price reversion towards its long run equilibrium. The particular specification is:

$$E(\Delta P_{t+1}^F) = -\alpha (P_t - F_t(X_t)) \quad \alpha > 0 \quad (5)$$

Where F_t is the fundamental price of the commodity (or the relevant commodity price index) in time t . This price is a function of a vector of variables (X_t) stressed by empirical literature which were discussed in section 2.3 of this paper.

According to equation (5), fundamentalists expect decaying (increasing) prices when current prices are higher (lower) than fundamental prices. Thus, they are willing to sell or buy the commodity in a counter-cyclical fashion. It is not necessary to assume that fundamentalists know exactly which the long term value of the commodity is. We can think instead they can obtain a consistent estimation of this equilibrium. For instance, it could be assumed that these agents have an imperfect knowledge about the real model because there exists uncertainty regarding the true value of the parameters, but they build their expectations based on an econometric regression without making systematic errors (Bray and Savin, 1986 and Fourgeaud et al., 1986).

Chartists, on the contrary, employ technical analysis and track the current trends in prices. One way to formalize this type of strategy is as follows:

$$E(\Delta P_{t+1}^C) = \delta (P_t - P_{t-1}) \quad \delta > 0 \quad (6)$$

¹¹ Pioneering work on heterogeneous agents literature corresponds to Frankel and Froot (1987a,b and 1990), DeLong et al. (1990a,b), and Shleifer and Summers (1990).

Every time price increases, these agents will take a long position in commodities because they expect that this trend will continue in the future. From the correlation analysis of Chapter 3 we know this is compatible with the actions of financial participants.

The key factor of this simple model is the inclusion of portfolio managers who are assumed to have an information advantage, in the sense they know the way the other market players form expectations. In order to take advantage of their knowledge, PM agents adjust their expectations employing a weighted average of expressions (5) and (6):

$$E(\Delta P_{t+1}^{PM}) = (1 - w_t)E(\Delta P_{t+1}^C) + w_t E(\Delta P_{t+1}^F) \quad 0 \leq w_t \leq 1 \quad (7)$$

It is crucial here the role of the variable w_t which governs the weight given to each type of expectation in time t . We assume that w_t adjusts endogenously in response to the size of past misalignment. Thus, the w_t variable is the source of non-linearity into the model. We propose, in particular, the following exponential function:

$$w_t = 1 - \exp\left(-\gamma \left(P_{t-d} - F_{t-d}(X_{t-d})\right)^2\right) \quad \gamma > 0, d \geq 0 \quad (8)$$

The intuition behind the specification of equation (8) is that the gap between actual and fundamental prices (lagged d periods) is the element that determines the weights assigned to the expectations of F and C agents by the portfolio managers. When the case is such that $F_{t-d} \cong P_{t-d}$, then w_t will show a very small value encouraging portfolio managers to follow the behavior of chartists. In the limit, when $F_{t-d} = P_{t-d}$ ($w_t = 0$), the price change will be given by:

$$\Delta P_{t+1} = (a_1 + a_3)E(\Delta P_{t+1}^C) = (a_1 + a_3)\delta(\Delta P_t) \quad (9)$$

If we are interested in empirically studying commodity price dynamics, expression (9) suggests employing a purely autoregressive econometric specification.

As long as the gap between F_{t-d} and P_{t-d} increases, portfolio managers start to bet against this misalignment. That is, the larger the misalignment is, the larger the weight they give to F expectations is. Again, in the extreme case $\exp\left(-\gamma \left(P_{t-d} - F_{t-d}(X_{t-d})\right)^2\right) \rightarrow 0$, and consequently $w_t = 1$ and $E(\Delta P_{t+1}^{PM}) = E(\Delta P_{t+1}^F)$. After some substitutions, the law of motion for the price dynamics will be given by:

$$\Delta P_{t+1} = a_1\delta(\Delta P_t) - (a_2 + a_3)\alpha(M_t) \quad (10)$$

Where we have defined $M_t = P_t - F(X_t)$.

The parameter γ is also relevant in this scheme, ruling the speed in which portfolio managers adjust their expectations through the w_t variable. If γ is quite high, for instance, even a small misalignment will induce PM agents to form expectations as the fundamentalists.

The general expression of commodity price changes can be obtained by replacing (5), (6), (7) and (8) into (4) and rearranging terms:

$$\Delta P_{t+1} = (a_1 + a_3)\delta\Delta P_t - a_2\alpha M_t - a_3\delta[1 - \exp(-\gamma M_{t-d}^2)]\Delta P_t - a_3\alpha[1 - \exp(-\gamma M_{t-d}^2)]M_t \quad (11)$$

Thus, price dynamics depends on several factors. The first two terms in equation (11) are based on a standard error correction model: a purely autoregressive term and a linear error correction factor. The remaining terms are those which generate the non-linear adjustment pattern. In the empirical analysis the emphasis will be placed on the non-linear adjustment coefficient of the price deviation from the long-run equilibrium.

Summarizing, we have developed a theoretical model in which fundamentals continue to be the only real force that drives long run prices. However, heterogeneity in expectations among market participants is important in determining the adjustment properties to equilibrium.

In the next section it will be shown that a specific type of smooth transition autoregressive model (STAR) tracks the commodity price dynamics derived from the model and described by the equation (11). With this econometric specification it will be feasible to study both the fundamental equilibrium and the potentially non-linear price adjustment pattern.

4.2. Econometric methodology

In the last decades there has been a growing interest in the use of non-linear econometric techniques. Among them, models with switching regimes turn out to be particularly attractive because they incorporate a law of motion that governs the shift from one state to another.

This law could be deterministic or stochastic. In the first case, the regime is determined by past values of observable variables, and it is known with certainty by all economic agents. In contrast, the state is stochastic if the regime is known only with some probability at every moment of time.

The simplest autoregressive model with deterministic regimes corresponds to the case of sudden changes and was developed by Tong (1978, 1990), Tsay (1989). In brief, it compares the transition variable (TV_t) with a threshold (c) in order to split up the linear model into two sub-models. If an autoregressive specification without explanatory variables is assumed, then the threshold model will be:

$$y_t = \begin{cases} (\phi_{1,0} + \phi_{1,1}y_{t-1} + \dots + \phi_{1,p}y_{t-p}) + \varepsilon_t & \forall y_t \leq c \\ (\phi_{2,0} + \phi_{2,1}y_{t-1} + \dots + \phi_{2,p}y_{t-p}) + \varepsilon_t & \forall y_t > c \end{cases} \quad (12)$$

In equation (12) switching between states is determined by the comparison between the transition variable (y_t) and the threshold (c), and it occurs abruptly.

The idea that the transition can be done gradually represents an important progress in this literature. It theoretically corresponds to the notion that economic agents do not react simultaneously when new information is spread or when a shock hits the economy. This empirical strategy is also valid if the effects of structural changes materialize slowly.

Moreover, whenever economic intuition or theory suggests that a relationship among variables is valid under certain circumstances but is no longer true if these circumstances change, then a smooth transition autoregressive (STAR) model will be suitable for its empirical test. Chan and Tong (1986); Granger and Teräsvirta (1993); Teräsvirta (1994), or Franses and van Dijk (2000) are pioneering references of this approach.

The representation of a smooth transition autoregressive model of p order or $STAR(p)$ is as follows:

$$y_t = (\phi_{1,0} + \phi_{1,1}y_{t-1} + \dots + \phi_{1,p}y_{t-p}) (1 - F(TV_{t-d}; \gamma, c)) + (\phi_{2,0} + \phi_{2,1}y_{t-1} + \dots + \phi_{2,p}y_{t-p}) F(TV_{t-d}; \gamma, c) + \varepsilon_t \quad (13)$$

Alternatively:

$$y_t = (\phi_{1,0} + \phi_{1,1}y_{t-1} + \dots + \phi_{1,p}y_{t-p}) + (\lambda_{2,0} + \lambda_{2,1}y_{t-1} + \dots + \lambda_{2,p}y_{t-p}) F(TV_{t-d}; \gamma, c) + \varepsilon_t \quad (14)$$

Where the following conditions are satisfied:

$$\lambda_{2,0} = \phi_{2,0} - \phi_{1,0}, \lambda_{2,1} = \phi_{2,1} - \phi_{1,1}, \dots, \lambda_{2,p} = \phi_{2,p} - \phi_{1,p} \quad (15)$$

$$E(\varepsilon_t | \Omega_{t-1}) = 0 \quad (16)$$

$$E(\varepsilon_t^2 | \Omega_{t-1}) = \sigma^2 \quad (17)$$

$$\Omega_{t-1} = (y_{t-1}, \dots, y_{t-p}) \quad (18)$$

In equations (13) and (14), the expression $F(TV_{t-d}; \gamma, c)$ is known as the transition function. It is a continuous function whose image is the interval $[0, 1]$.¹² The γ parameter measures how fast the adjustment between regimes is, while the parameter c , establishes the limit point after which the switching between regimes start to take place. Finally, TV_{t-d} refers to the transition variable with d lags.

There are very few technical restrictions on the sort of variable(s) that could be TV_{t-d} . Usual options include lags of endogenous variables, exogenous variables, functions of endogenous and/or exogenous variables or a time trend (van Dijk et al., 2002).

On some occasions, however, a theoretical hypothesis suggests which variable can be regarded as a determinant of the transition. This is precisely the case of commodity prices and their possible pattern of non-linear equilibrium correction. From the heterogeneous agent model we obtain a specific theoretical restriction establishing that past misalignment between the current price and the fundamental value not only acts as a determinant of price changes but also as the variable that governs the transition between regimes.

Usual TV_{t-d} functions are either logistic or exponential:

$$F(TV_{t-d}; \gamma, c) = \frac{1}{1 + \exp(-\gamma(TV_{t-d} - c))}, \gamma > 0 \quad (19)$$

¹² Both properties differentiate a smooth transition autoregressive model from a threshold model, because in the latter the transition function is direct, taking only two values: 0 or 1.

$$F(TV_{t-d}; \gamma, c) = 1 - \exp\left(-\gamma(TV_{t-d} - c)^2\right), \gamma > 0 \quad (20)$$

The logistic function¹³ allows us to distinguish between two regimes or states, named high and low regimes respectively. The high state arises from positive and large values of $(TV_{t-d} - c)$, since $\exp\left(-\gamma(TV_{t-d} - c)^2\right)$ tends to zero, and hence the expression (19) tends to 1. By contrast, the regime is low when $(TV_{t-d} - c)$ takes low values and so $\exp\left(-\gamma(TV_{t-d} - c)^2\right) \rightarrow \infty$ y $F(TV_{t-d}; \gamma, c) \rightarrow 0$.¹⁴ The logistic specification is valid when it is believed that the transition takes place in a monotonic way.

On the contrary, the exponential function is useful if the value of absolute deviation of the transition variable with respect to parameter c is the important feature. This specification is known as ESTAR model (exponential smooth transition autoregressive model) and restricts the dynamics of the equation to be the same alongside the extreme values of $F(TV_{t-d}; \gamma, c)$.

Thus, in an ESTAR model the asymmetries between regimes are given by the absolute magnitude of the differences rather than by their sign.

Some well-known applications of STAR models could be found in papers studying misalignment of the real exchange rate regarding its fundamental value given by the purchasing power parity (Michael et al., 1997; Taylor et al., 2001, or Chen and Wu, 2000); in the literature of non-linear adjustment of deviations from uncovered interest rate parity (Sarno et al., 2006); or in those works that test non-linear mean reversion in stock futures (Monoyios and Sarno, 2002).

The extensions of STAR models to multivariate contexts (VAR models or systems of equations) have been studied, among others, by Weise (1999), van Dijk (2001), Camacho (2004) and Mendoza (2004). The general structure of a non-linear equilibrium correction model is:

$$\Delta Y_t = \Pi_{1,0} + \Gamma D_t + \alpha_1 M_{t-1} + \sum_{j=1}^p \Pi_{1,j} \Delta Y_{t-j} + \left(\Pi_{2,0} + \alpha_2 M_{t-1} + \sum_{j=1}^p \Pi_{2,j} \Delta Y_{t-j} \right) F(TV_{t-d}; \gamma, c) + \varepsilon_t \quad (21)$$

Where ΔY_t is a $(n \times 1)$ vector, $(\Delta y_1, \Delta y_2, \dots, \Delta y_n)$, D_t is a $(m \times 1)$ vector of dummies which control for outliers and Γ $(m \times n)$ is the respective matrix of coefficients. Also, $\Pi_{1,0}$ and $\Pi_{2,0}$ are $(n \times 1)$ vectors containing the constants of the linear and non-linear part in each case. The $\Pi_{1,j}$ and $\Pi_{2,j}$ are $(n \times n)$ matrices for $j: 1, \dots, p$ that correspond to the autoregressive coefficients. The equilibrium correction term is denoted by $M_t = \beta' X_t$, where β is the $(n \times r)$ matrix of coefficients of the long-term relationship and X_t is a $(n \times 1)$ vector which stands for the variables in levels. Finally, α_1 and α_2 are $(n \times r)$ matrices formed by the adjustment coefficients of deviations from long-term relationships, where r indicates the number of cointegration equations. These coefficients play a fundamental role in the model since they capture the linear and non-linear adjustment pattern.

¹³ A logistic smooth transition autoregressive (LSTAR) model is obtained when function (17) is applied.

¹⁴ Since the transition function can take any continuous values between zero and one, characterization of a STAR model with only two regimes may look quite arbitrary, particularly in those cases in which the smoothing parameter is low and, therefore, there are many intermediate values of the transition function. In this sense, van Dijk et al. (2002) argue that the STAR model can be thought of as a methodology that allows a "continuous" set of regimes.

Concerning the operational steps needed to implement a STAR model, Teräsvirta (1994) proposes a procedure for the univariate case, whereas Granger and Teräsvirta (1993) examine the multivariate case and Camacho (2004), among others, extends it to a multi-equational approach. In all cases, the process encompasses basically four stages: i) estimating a linear model ii) testing non-linearity; iii) estimating the non-linear model, and iv) computing non-linear impulse-response analysis.

The first step is to estimate a linear model that will serve as a benchmark to contrast the non-linearity hypothesis. The estimation sequence follows common techniques of time series analysis. In this point, it is important to control for outliers and to check the behavior of the residuals regarding autocorrelation and heteroscedasticity.

The second stage consists of a linearity test. If the alternative hypothesis to the linearity is the smooth transition exponential model (STAR),¹⁵ Teräsvirta (1994) suggests using a first order Taylor expansion (T_1^e) to obtain an auxiliary regression which may serve as a base to contrast the null hypothesis. To this end, one should take equation (20) and calculate T_1^e in $\gamma = 0$. After some simplifications, the following auxiliary regression will be achieved (vector notation):

$$Y_t = \beta_0' X_t + \beta_1' X_t TV_{t-d} + \beta_2' X_t TV_{t-d}^2 + \omega_t \quad (22)$$

Where $X_t = [1, y_{t-1}, y_{t-2}, \dots, y_{t-d}]$; $\beta_0 = [\beta_{0,0}, \beta_{0,1}, \dots, \beta_{0,p}]$; $\beta_1 = [\beta_{1,0}, \beta_{1,1}, \dots, \beta_{1,p}]$; and $\beta_2 = [\beta_{2,0}, \beta_{2,1}, \dots, \beta_{2,p}]$
In (20), testing the linearity hypothesis is equivalent to prove that $\beta_1 = 0$ and $\beta_2 = 0$.

According to Teräsvirta (1994), the recommended procedure consists of an F -test with the following sequence: i) estimate the model under the assumption of linearity and compute the residual sum square (RSS_0); ii) estimate the auxiliary regression (22) to obtain RSS_1 ; and iii) compute the critical value of the LM statistics:

$$LM = \frac{(RSS_0 - RSS_1) / l_1}{RSS_0 / l_2} \quad (23)$$

The l_1 degrees of freedom are calculated as the difference between the number of parameters in the unrestricted and restricted models, whereas l_2 is calculated as the number of observations minus the unrestricted model parameters. If the linearity test is performed on a list of possible transition variables, it will be necessary to define which variables will be considered. In such a case the highest LM statistics is employed to select the transition variable.

The third step is the estimation of the STAR model, which can be done by any conventional non-linear method. This will require the definition of initial conditions. An appropriate selection of these conditions will increase the probability of reaching a maximum in the likelihood function.

¹⁵ The auxiliary regression is modified when the specification is done throughout a logistic function. There is also evidence that expression (20) is appropriate when there is no knowledge (or no prior intuition) whether the relevant alternative is the expression (17) or (18). In this regard, see Luukkonen et al. (1998).

The usual practice to find initial conditions is conducting a two dimensional grid search on parameters γ and c . It is important to note in this regard that once the values of both parameters are fixed, the function $F(TV_{t-d}; \gamma, c)$ will lie in the interval $[0, 1]$ in each observation, being equation (13) linear in all its arguments. The grid search iterates different values of γ and c taken at intervals that are relevant in accordance with their respective scales of variability. The conditional estimation can be done by a linear method as OLS or SUR. The configuration of (γ, c) which generates the restricted model with the maximum likelihood will be selected. Then, its parameters are used as initial conditions to estimate the unrestricted STAR model.

Theory could also provide some conditions for the smoothing parameter (γ) or the threshold (c). In those models where the misalignment is the state variable ($TV_{t-d} = M_{t-d}$), the condition $c = 0$ is often imposed because of the symmetry in the economic concept of misalignment itself. Thus, when the differences between actual and fundamental prices are very small, then $F(TV_{t-d}; \gamma, c=0) = 1 - \exp(-\gamma(M_{t-d})^2) \rightarrow 0$, and so we will observe the system operating in one of the extreme regimes. In papers about PPP theory, transaction and transportation costs limit arbitrage and thus small mismatches are interpreted as a state variable, where PPP does not hold. The corollary is that the exchange rate behaves like a random walk if (M_{t-d}) is low.

Finally, a useful tool of a STAR model (available only in equation system case) is the computation of generalized impulse-response functions.

A methodological description of these impulse-response functions is beyond the scope of this paper. However, the intuitive idea is that in a STAR model the shock effects depend on the history, size and sign of disturbances. For instance, the effect of a shock will not be necessarily the same if the shock occurs when there is a small misalignment with respect to the case of a high initial gap between current and equilibrium commodity prices. Also the size of shock could be relevant since it could involve different future dynamic trajectories of the endogenous variables. These characteristics are not incorporated by traditional impulse-response functions estimation methods.

We follow the bootstrapping methodology of Koop et al. (1996) to compute the generalized non-linear impulse-response functions. We suggest reviewing this reference for a complete discussion of technical details.

5. Empirical evidence: Non-linearity in the adjustment of commodity prices

With the aim of organizing the presentation of the empirical model, this chapter has been divided into three parts. The structure of the model to be estimated is described in the first part. In the second one, we present the non-linearity tests results, the interpretation of the transition function and different regimes, and the estimation of the non-linear system. Finally, impulse-response analysis is performed to investigate the short-term reaction of commodity prices to shocks in fundamentals under both high and low misalignment regimes.

5.1. The structure of the short run empirical model

The structure of the model to be estimated is similar to a VECM but includes non-linear terms in the commodity price equation. In this sense, we move away from the seminal work of Weize (1999) that specifies a symmetrical smooth transition VAR; which means all the equations present non-linear terms. Our decision

is based on the fact we only have a theoretical rationale for non-linear adjustment in the price dynamics. An alternative empirical strategy could have been using a single-equation STAR.¹⁶ However, that option would have implied abandoning the possibility of studying the short-term commodity price response to shocks in fundamentals and we would not be able to study interactive effects.

In particular, we take (21) as a benchmark and work with the following system:

$$\Delta P_t = \beta_{11} + \alpha_{11}(M_{t-1}) + \sum_{j=1}^p \Pi_{11,j} \Delta X_{t-p} + \left(\beta_{12} + \alpha_{12}(M_{t-1}) + \sum_{j=1}^p \Pi_{12,j} \Delta X_{t-p} \right) (1 - \exp(-\gamma TV_{t-d}^2)) + \varepsilon_t^P \quad (24)$$

$$\Delta RER_t = \beta_2 + \alpha_2(M_{t-1}) + \sum_{j=1}^p \Pi_{2,j} \Delta X_{t-p} + \varepsilon_t^{RER} \quad (25)$$

$$\Delta IR_t = \beta_3 + \alpha_3(M_{t-1}) + \sum_{j=1}^p \Pi_{3,j} \Delta X_{t-p} + \varepsilon_t^{IR} \quad (26)$$

$$\Delta PRO_t = \beta_4 + \alpha_4(M_{t-1}) + \sum_{j=1}^p \Pi_{4,j} \Delta X_{t-p} + \varepsilon_t^{PRO} \quad (27)$$

$$\Delta DOW_t = \beta_5 + \alpha_5(M_{t-1}) + \sum_{j=1}^p \Pi_{5,j} \Delta X_{t-p} + \varepsilon_t^{DOW} \quad (28)$$

Where $X = [P, RER, IR, PRO, DOW]$ and the threshold coefficient c is set equal to zero because of theoretical reasons discussed in section 4.2. As before, $M_t = P_t - F(X_t)$ and TV_{t-d} is the transition variable. The main focus of the empirical analysis is centered on the equation (24) as it contains the price dynamics which is similar to that derived from the theoretical model (equation (11)). Particular interest should be given to the estimates of α_{11} and α_{12} which represent the price adjustment coefficients to the deviations from the long-term equilibrium.

The econometric strategy adopted is in line with the Engle and Granger proposal. In essence, we estimate in a first stage the long run equation for commodity prices and test for cointegration. Then, if a cointegration relationship is found, the whole error correction system will be estimated using $M_t = P_t - F(X_t)$ as a regressor.

We have already conducted the first step of this strategy in the section 2.3. The second stage is carried out in the next section.

¹⁶ This is the strategy adopted by Westerhoff and Reitz (2005) to explain mean reversion in corn prices. Other difference of that model in comparison to our empirical framework is that these authors do not use an exponential specification for the transition function.

5.2. Econometric estimation of the short-term model and nonlinear equilibrium adjustment

As it was stated in section 4.2, testing the nonlinear equilibrium adjustment hypothesis requires estimating a linear model which acts as benchmark.

To this end, we specify a linear symmetrical vector error correction model (VECM) using the previously calculated misalignment. The Akaike Information Criterion (AIC) suggests using five lags for the changes in the variables. In addition, under this specification the residuals do not present autocorrelation.¹⁷

The adjustment coefficient (α_{11} in equation (24)) takes a value of -0.01921 and is statistically significant at 10% when Newey-West HAC standard errors are considered. Thus, there is evidence that, at least in average, correcting forces emerge when commodity prices are in disequilibrium.

However, as our theoretical model of section 4.1 predicts, it is really possible that there exist some states where misalignments are high and hence correcting forces operate, while in other states characterized by low misalignments the gaps remain. This is the intuition behind the non-linear equilibrium adjustment hypothesis.

Once we have estimated the benchmark linear model, we carry out the linearity F-test on the commodity price equation as it was described in the methodological section 4.2. The auxiliary regression takes the specific following form:

$$\begin{aligned} \Delta P_t = & \beta'_{0,p} \Delta X_{t-p} + \nu_1 M_{t-1} + \beta'_{1,p} \Delta X_{t-p} TV_{t-d} + \nu_2 M_{t-1} TV_{t-d} + \\ & \beta'_{2,p} \Delta X_{t-p} TV_{t-d}^2 + \nu_3 M_{t-1} TV_{t-d}^2 + \omega_t \end{aligned} \quad (29)$$

Where $\Delta X = [1, \Delta P_{t-p}, \Delta RER_{t-p}, \Delta IR_{t-p}, \Delta PRO_{t-p}, \Delta DOW_{t-p}]$, TV_{t-d} stands for the transition variable, and $p : 1, \dots, 5$. Note that the first two terms in equation (29) correspond to the linear specification of the commodity price equation (24) in the VECM model described above.

As we have already stressed, our theoretical model suggests the regimes are governed by the size of misalignment. Thus, we propose five potentially transition variables. One of them is simply the misalignment (M_{t-d}) and the others result from averaging out the current misalignment and those of the previous j periods (AVj_{t-d}), where j ranges from 1 to 4. Besides, we consider, $d = 1, \dots, 12$ lags for each of them.¹⁸

Table 3 shows the test results for the ten variables which exhibit the lowest p-values.¹⁹ Remember that the null hypothesis is linear adjustment in the commodity price equation (24).

We find evidence of non-linearity in equation (24) at 5% statically significance when the variable $AV1_{t-1}$ is considered. That is, regime in each moment t would be defined by the average value of the misalignments registered in $t - 1$ and $t - 2$.

¹⁷ We performed the Serial Correlation LM test with 12 lags and we failed to reject the null hypothesis of no autocorrelation at 5% statistical significance.

¹⁸ We do not test $TV_{t-1} = M_{t-1}$ because of perfect multicollinearity problem when system (24)-(28) is estimated.

¹⁹ This test was programmed on E-views. The code is available upon request.

Table 3

Linearity F-test results

Transition Variable	LAG	Statistic LM	p-value
AV1	1	1.38647	0.04912
AV4	12	1.38309	0.05040
AV3	12	1.21178	0.16396
AV4	11	1.19485	0.18166
AV1	6	1.15964	0.22287
AV2	1	1.15512	0.22859
AV2	12	1.12511	0.26910
AV2	6	1.12423	0.27036
AV4	4	1.12396	0.27074
AV3	5	1.12106	0.27490

Therefore, we will estimate the non-linear system (24)-(28) considering $AV1_{t-1}$ as transition variable. In the section 4.2 we have anticipated which is necessary for this purpose to provide proper initial conditions for all the coefficients of the system.

Those conditions can be found estimating the linear system which is obtained after fixing the value of the parameter γ in equation (24). Instead of carrying out a grid search, we select a value of γ such that the transition function averages 0.5 in the whole sample. Intuitively, this means that portfolio managers assign in average the same weight to both fundamentalist and chartist agent expectations. Consequently, initial value of γ approximately should satisfy the following expression:

$$\bar{w}_t = \frac{1}{T} \sum_{t=1}^T \{1 - \exp(-\gamma AV1_{t-1}^2)\} \cong \frac{1}{2} \quad (30)$$

We find that $\gamma = 133.77$ numerically solves (30) for the 1973-1 to 2008-5 period. Then, after setting γ at this value, the system (24)-(28) is estimated by SUR and initials conditions for all the coefficients are obtained.

After finding these conditions, we proceed estimating the unrestricted non-linear system also by SUR.²⁰ In order to evaluate our hypothesis, analysis is centered in the adjustment coefficients of the commodity price equation, namely α_{11} and α_{12} .

The linear error correction coefficient α_{11} appears not statistically significant (p-value=0.2914) and takes a positive value of 0.035. Contrary, the non-linear adjustment coefficient α_{12} results significant at 5% (p-value=0.0418) and is equal to -0.109. Moreover, the γ parameter reaches a magnitude of 55.22 with a p-value equal to 0.0292.

These estimations support our non-linear adjustment hypothesis due to heterogeneous agents in commodity markets.²¹ To see this, we define the Global Equilibrium Correction Factor (*GEC*) as the two terms of the price equation (24) containing the price deviation from its long run equilibrium (M_{t-1}).

²⁰ The E-Views code used in this section is also available upon request to the authors.

²¹ Most of the arguments will be stated considering positive misalignments, but the same applies for negative gaps because of the symmetry of exponential function.

$$FCGE = (0.035)M_{t-1} + (-0.109)M_{t-1} * w_t = \{(0.035) + (-0.109)w_t\} M_{t-1} \quad (31)$$

When misalignment is low enough (low regime), w_t tends to zero and the second term of (31) vanishes. In this case, portfolio managers mimic chartist investors and initial misalignment is widened at a 3.5% monthly rate. Under these circumstances, there is no equilibrium correction.

However, when the gap between actual and equilibrium prices reach a sizable value (high regime), w_t tends to one and the *GEC* adjustment coefficient attains a maximum of -0.074 ($=\alpha_{11} + \alpha_{12}$). Intuitively, portfolio managers assign a larger weight to fundamentalist investor expectations when misalignment is high and therefore we will observe price reversion toward the equilibrium.²²

A key question is to determine what high and low commodity price misalignment means in our model. Figure 16 provides information to answer it.

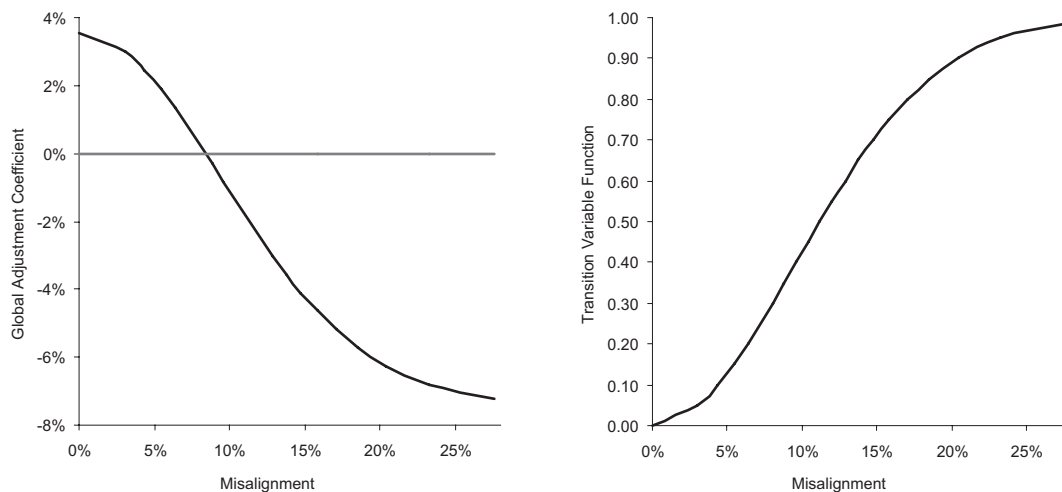
The LHS panel of Figure 16 reveals that price correction forces prevail in the market only when gap is larger than 8.5%. Considering the transition variable function (RHS panel), adjustment to equilibrium predominates when TVF (and w_t) exceeds 0.32. Furthermore, both panels show that non-linearity implies bigger adjustment coefficients, the higher the misalignment is. Maximum reversion speed (-7.4% monthly) is attained when spread between actual and fundamental commodity prices surpasses 25%.

The corollary is that higher misalignments in the past (measured by AVI_{t-1}) involve higher values of the transition function and this will indicate stronger future trend reversals.

Finally, Figure 17 depicts the transition function in the sample considered.

Figure 16

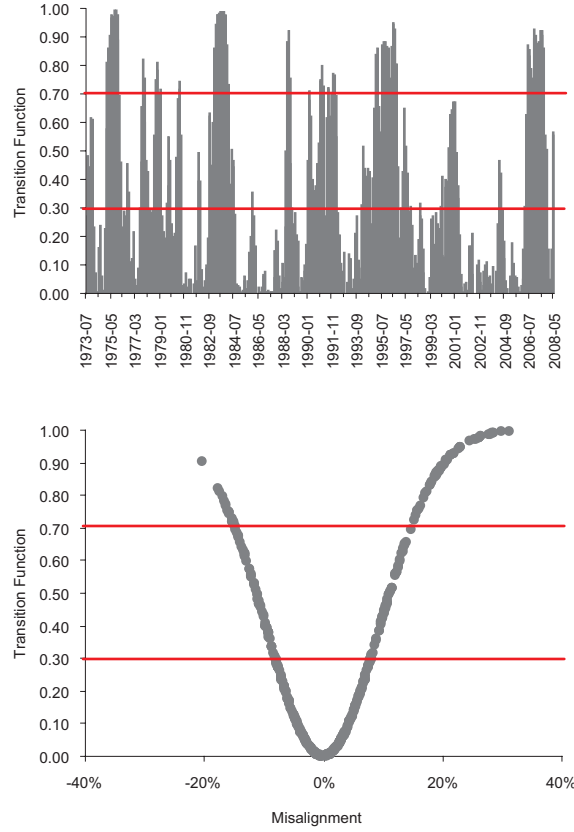
Misalignment, transition function and global adjustment coefficient



²² Given that non-linear adjustment coefficient (α_{12}) is negative and larger than α_{11} in absolute terms, the overall stability of the price equation is guaranteed.

Figure 17

Transition function in the 1973-1/2008-5 sample



In order to analyze the distribution of the states, we will assign observations to low regime if transition function takes values lower than 0.30 whereas we will consider they belong to high regime every time transition function exceeds 0.70.²³ The remaining observations form the “transition regime”. Given this criterion, we find 56% of the time the market is dominated by chartists while fundamentalists only prevail about 18.85%. The rest of the time (25%) corresponds to transition periods.

Transition function averages out 0.34 in the whole sample. Therefore, mean price change expected by portfolio managers is as follows:

$$E(\Delta P_{t+1}^{PM}) = (1 - \bar{w}_t)E(\Delta P_{t+1}^C) + \bar{w}_t E(\Delta P_{t+1}^F) = 0.66E(\Delta P_{t+1}^C) + 0.34E(\Delta P_{t+1}^F) \quad (32)$$

The *GEC* coefficient associated to this figure is just -0.18%.

Summarizing, results support the hypothesis that high discrepancies between actual and equilibrium prices tend to be corrected relatively fast, while small misalignments tend to persist over time without any endogenous correcting force taking place.

In the next section we perform impulse-response analysis following the theoretical guidelines exposed in the fourth section.

²³ These figures imply that global adjustment coefficient is positive in the low state and larger in absolute value than -4% in the high state.

5.3. Non-linear impulse-response analysis

As it was stressed in the methodological section, in an exponential STAR model the effects of shocks depend on the history and size of disturbances.²⁴ These properties are not incorporated into traditional impulse-response analysis.

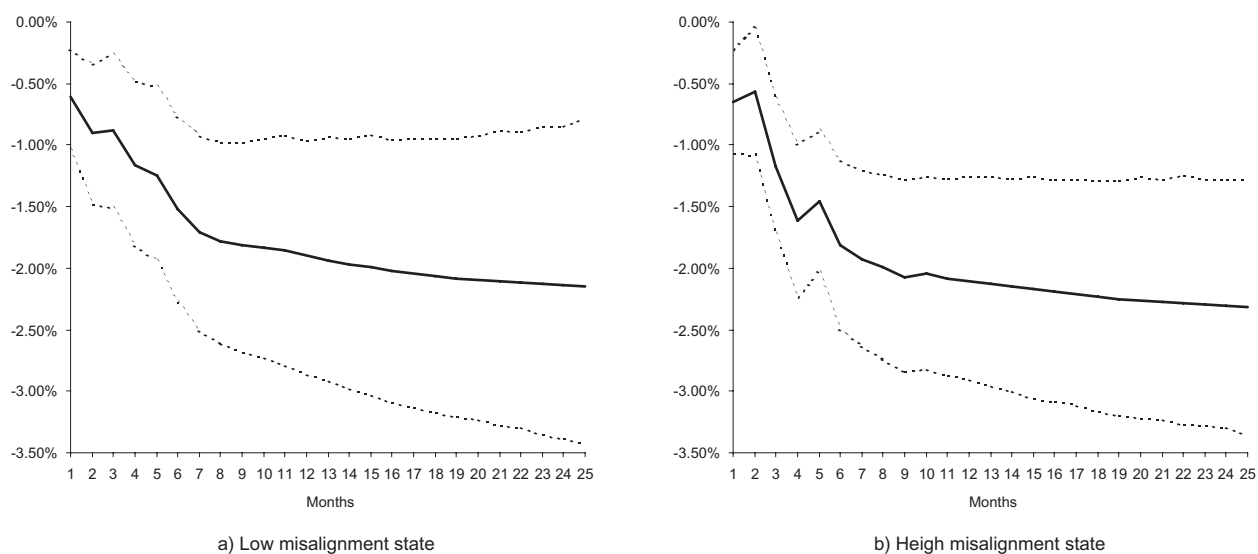
Before presenting the results we will indicate two technical issues. In the first place, to carry out the generalized impulse-response function of Koop et al. (1996) requires assigning each observation to different regimes. We do this adopting the same allocation criterion stated in the previous section. Then, sampling of shocks is performed.

Secondly, initial state is defined by the analyst but after shock takes place, it could endogenously change as consequence of the system dynamics.

Figures 18 to 21 show the accumulated change of commodity prices to one-standard deviation shock in the fundamentals for both high and low initial misalignment states. In the Appendix the effects of three-standard deviation shocks are depicted. The dashed lines are the 10% confidence bands.

Figure 18

US effective real exchange rate shock



²⁴ This is the case when the transition function is an exponential function. If we had selected a logistic specification, the sign of the shock would also matter.

Figure 19
Industrial production shock

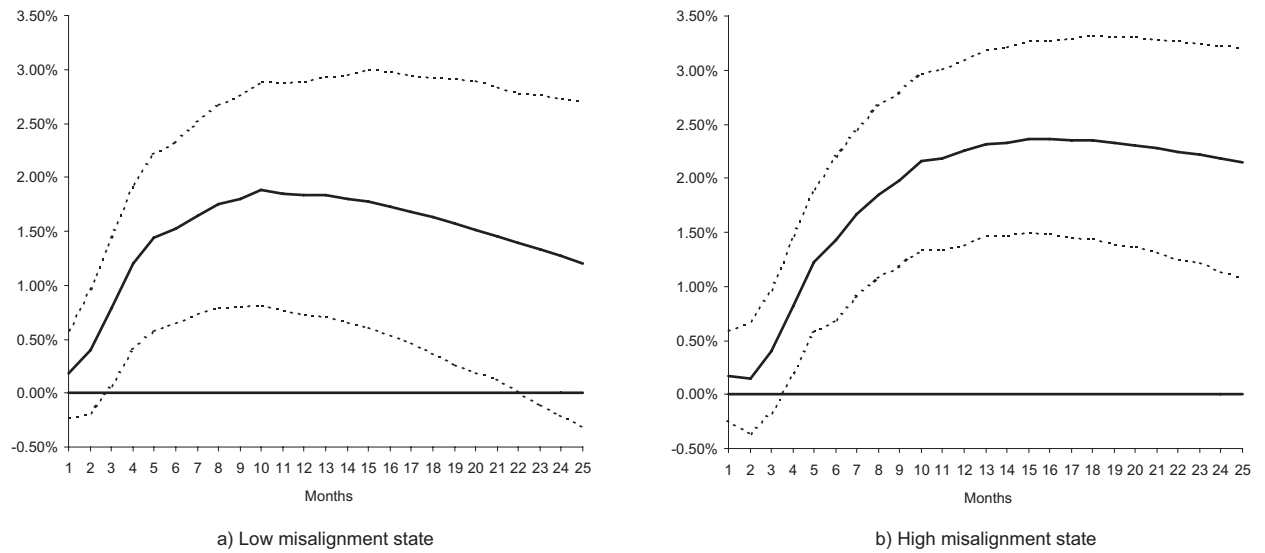


Figure 20
US real interest rate shock

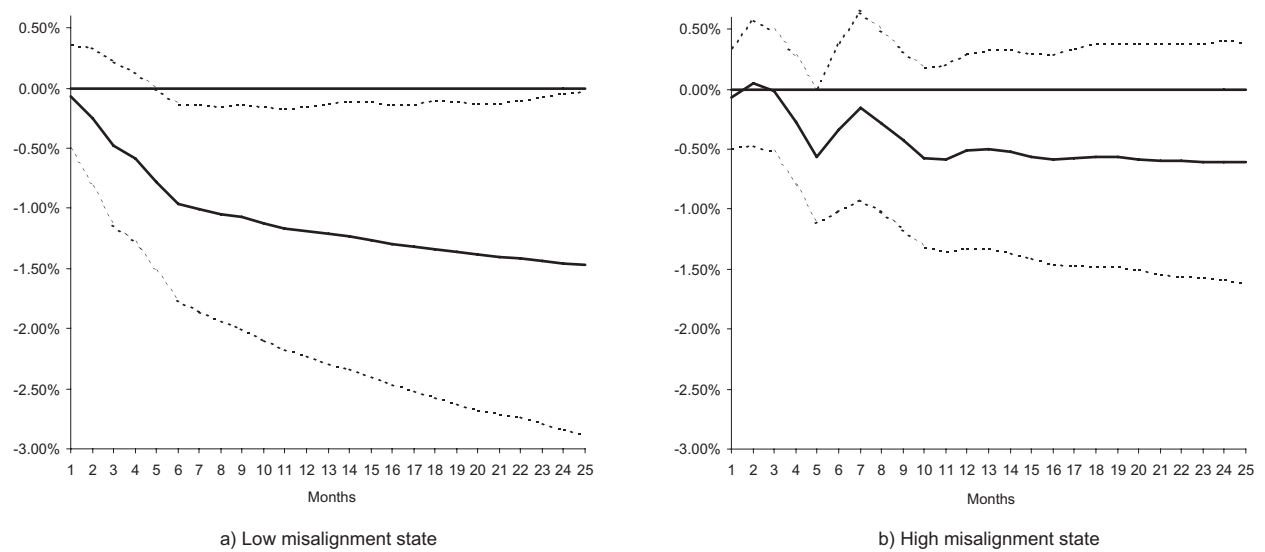
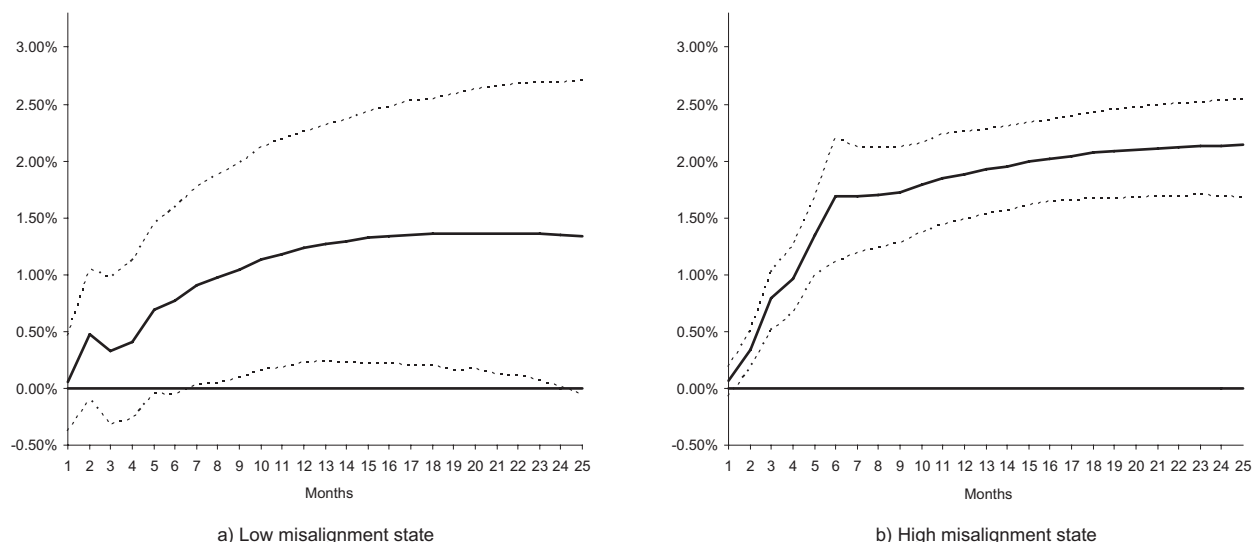


Figure 21

Real Dow Jones index



The main conclusion we could draw from the figures is that, in general, the responses obtained are in line with both the theoretical predictions and the long-run relationship estimation. The only exception is that we find a positive impact of the Dow Jones index in the short run.

Furthermore, it is noteworthy that price dynamics responses do not exhibit significant differences either the initial state is low or high. The effect of real interest rate shock differs from this general picture. If the shock occurs in the high regime, the commodity price reaction is minor and statistically non significant.

The same remarks apply when we study the consequences of bigger shocks (three-standard deviation). There are not important discrepancies when the effects of small and big shocks are compared, except for the interest rate shock response which becomes statistically significant whatever the initial regime is.

6. Open debates: Recent structural changes

As stated in the introduction, this section will deal with two processes whose origins back to 30 years ago and they have intensified in the last decade: the production of biofuels and the change in the pattern of international demand of primary goods caused by the structural transformation of China and India, two emerging countries with a heavy influence on the world economy. These processes must be taken into account when our empirical results are used to shape future scenarios for commodity markets.

Although, to some extent, the quantitative effect of these phenomena is captured in the empirical model through the variable representing the international demand, there may be doubts about its appropriateness to measure the real dimension of the impact of these processes on the long-term trend of commodity prices. Consequently, this section provides additional assessment criteria contributing to a more comprehensive view of the commodity market's future outlook.

6.1. Biofuels

As pointed out by the report prepared by the United Nations Environment Program (UNEP, 2009), climate change, the increasing demand of energy and the sharp volatility of oil prices have encouraged the search for alternative sources of energy with three attributes: economic efficiency, social equity and environmental sustainability. Among the options developed, biofuels have generated a remarkable interest both in public and private sectors.

The universe of biofuels encompasses the materials used as energy sources that derive, either directly or indirectly, from biomass. Biofuels may be classified as solid (e.g. firewood), liquid (such as biodiesel and ethanol) and gaseous (e.g. biogas).

Biofuels for transportation, together with the biomass processed for heating and electrical power generation, are part of what is known as modern bioenergy. According to their current or future availability, they are grouped in first-, second- and third-generation biofuels.

First-generation biofuels are commercially produced with conventional technology and are obtained from agricultural crops traditionally used for human consumption and animal feeding. In this group, ethanol and biodiesel are the two leading biofuels both in traded volume and produced value.

Since they are made from raw materials such as corn, soybean or sugarcane, ethanol and biodiesel production is especially relevant because of its potential direct impact on commodity prices.

Basically, ethanol can be obtained from any raw material with high sugar content (sugarcane, sugar beet) or, otherwise, rich in starch or cellulose (corn, wheat). In turn, biodiesel is the result of a chemical process known as transesterification, based on vegetable oils from crops such as soybean in Argentina, Brazil and the United States, rape in Europe or palm oil in several South-East Asian countries (Indonesia, Malaysia). The combination of ethanol with gasoline, and biodiesel with diesel, both result in a better combustion since it reduces carbon monoxide and sulfur oxide emissions.

Although liquid biofuels have already been produced for several years (the Brazilian *Proálcool* Program of 1975 is an example), they still account for a very small percentage of fuel total demand for transportation (1.8% in 2007) and of energy global consumption (0.3% in 2006 against 79% for fossil fuels in that same year).

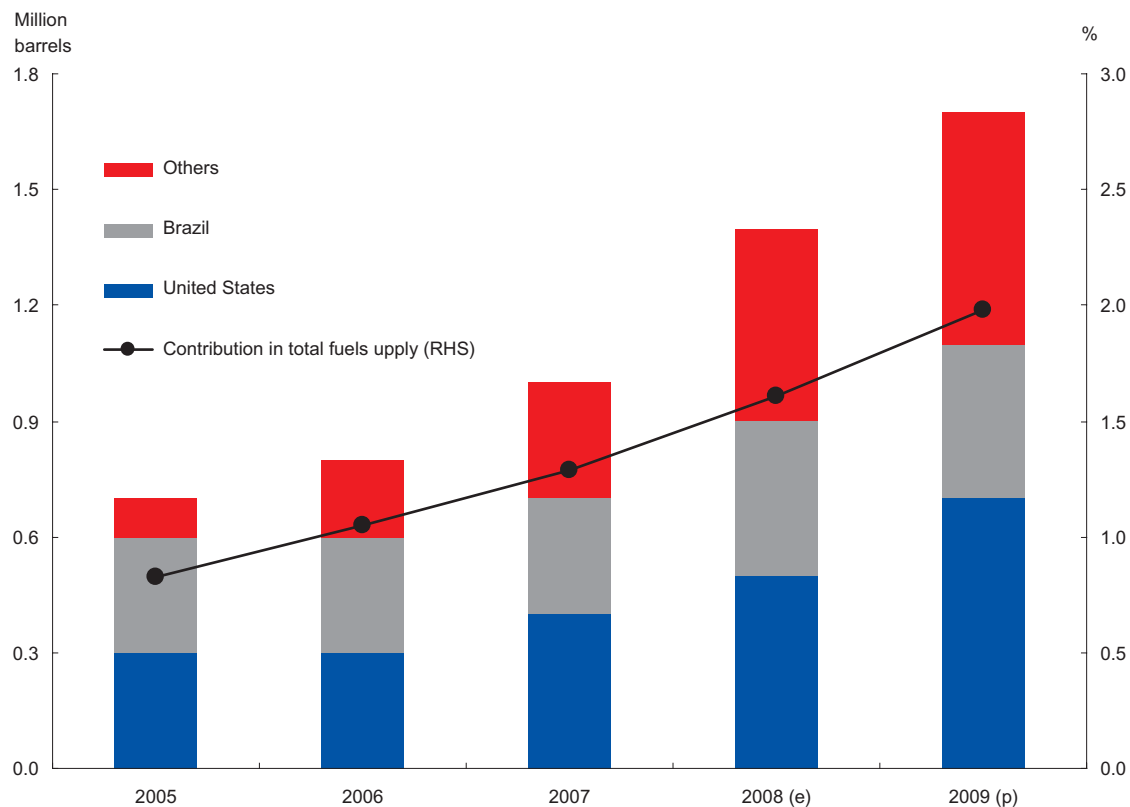
However, since the early 2000s, the generation of biofuels has experienced a remarkable growth. Ethanol production four-folded between 2000 and 2009, from 17 billion liters to over 74 billion liters, while in the same period the increase in the amount of biodiesel produced was even more impressive, from around 1 billion liters to over 14 billion liters (FAPRI, 2010).

In turn, Figure 22 shows that in terms of equivalent oil barrels, the share of biofuels in the total supply of fuels grew by 132% in the 2005-2009 period, accounting for nearly 2%.

At present, the production of ethanol clearly surpasses other liquid biofuels, representing approximately 85% of the total, while biodiesel accounts for the remaining 15%.

Figure 22

World production of biofuels (equivalent oil barrels per day)



Source: Author's calculations based on data from the International Energy Agency (IEA).

The United States and Brazil are leaders in ethanol production worldwide, generating 53.4% and 32.7%, respectively (Figure 23). These countries have followed different strategies, since in the case of the United States the raw material is corn while Brazil uses sugarcane.

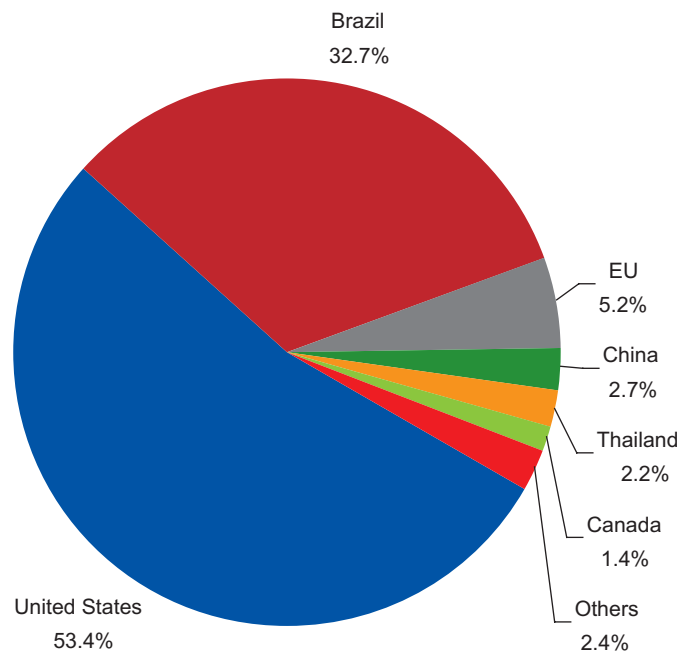
Regarding biodiesel world production, the European Union captures around 50%, on the basis of rapeseed oil, followed by the United States (11%), Brazil (10.4%) and Argentina (8.4%), though these countries use soybean oil as basic raw material (Figure 24). In turn, Thailand (3.4%) produce biodiesel from palm seed oil, the same as other South-East Asian countries like Indonesia and Malaysia.

The increasing interest in the production and use of ethanol and biodiesel as renewable fuels to replace or supplement gasoline and gas oil results from a combination of factors such as the increase in oil price in international markets, the search for a more diversified energy matrix and the current pressures to reduce carbon dioxide emissions to mitigate the climate change.

Thus, in recent years, an increasing number of countries, both developed and developing, have established minimum percentages of biofuels that must be incorporated to gasoline and gas oil for automotive transportation, such as the United States, Canada, France, Germany, Brazil, China and Argentina (Goldemberg et al., 2008; Mathews, 2007).

Figure 23

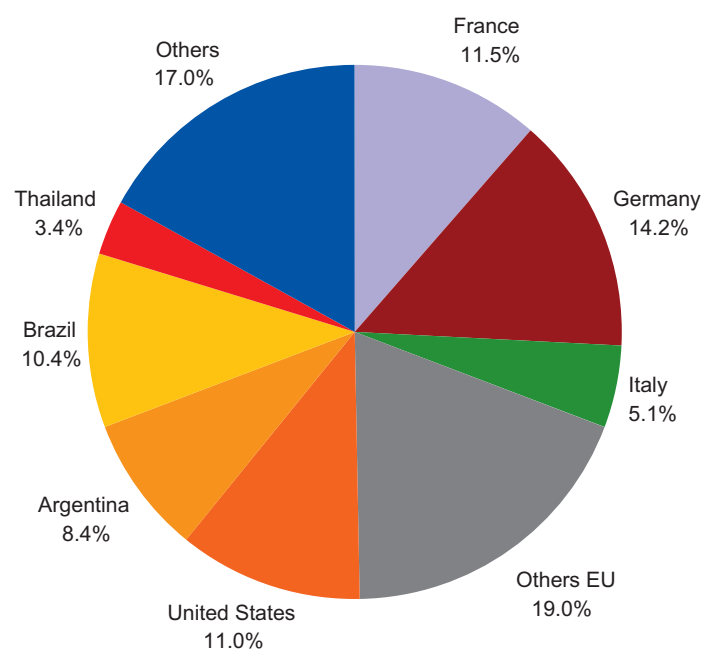
Ethanol production worldwide (2009)



Source: Author's calculations based on data from the Renewable Fuels Association.

Figure 24

Biodiesel production worldwide (estimated for 2010)



Source: Author's calculations based on data from World Oil.

In addition, incentive policies for promoting the production and consumption of biofuels have been implemented, given the reduced economic feasibility of some productive schemes.

The governmental support through subsidies, tax exemptions and benefits may be present at any point of the production and commercialization chain. This means that they can be granted in the early production stage, through the application of specific policies to raw materials used in the process (water, crops and energy) or in the refining and storage stages, by means of fiscal incentives, also providing support for the construction of infrastructure. Finally, in relation to the commercialization of biofuels and their byproducts, governments play a key role through official regulations on the composition of biofuels and tax exemptions for the purchase of flexible vehicles using both types of fuels. In addition, there is also a contribution by the public sector through research and development activities. Figure 25 summarizes the main incentive schemes in the different stages.

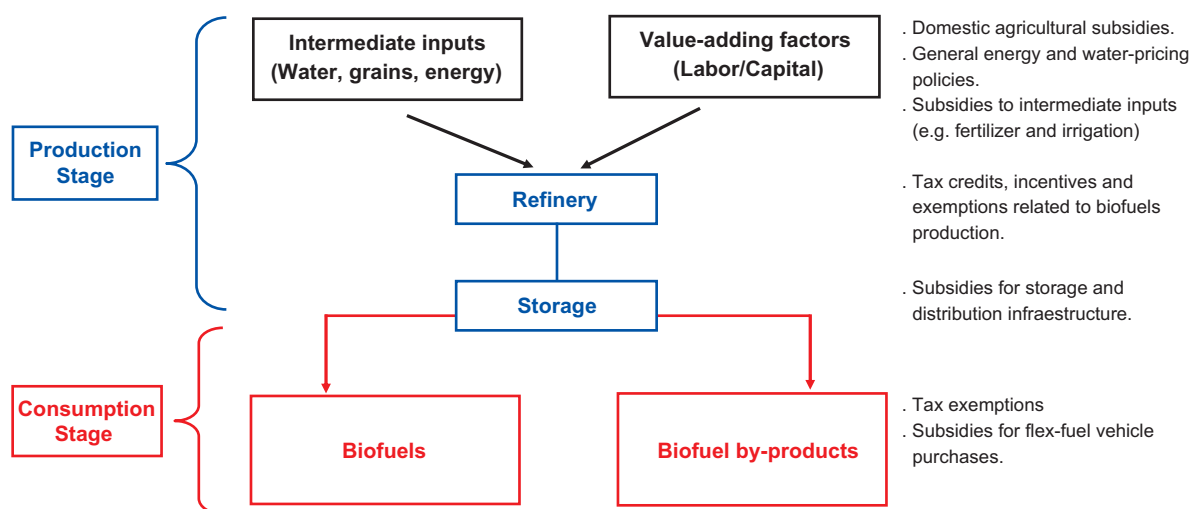
Although the assessment of comprehensive costs of subsidies directly or indirectly associated with biofuel production is a difficult task to perform in practice, a series of studies have been made to this effect for some OECD member countries.

According to Steenblik (2007), the total estimated governmental support to production and consumption of ethanol and biodiesel reached USD 10.8 and USD 12.1 billion, respectively, for a surveyed group of countries in 2006 (Table 4). However, the author points out that these values may be underestimated due to the difficulty to accurately measure investment incentives, mainly in the United States and the European Union.

Tyner and Taheripour (2007) estimate the alternative combinations of corn and crude oil prices for which ethanol production in the United States is profitable. This document is relevant since it rules out tax incentives and strictly focuses on the prices of both commodities to determine whether it is convenient to produce this biofuel.

Figure 25

Governmental incentive schemes in the biofuels productive chain



Source: Author's calculations base on Steenblik (2007).

Based on Tyner and Taheripour (2007) and FAO (2008), Figure 26 illustrates the indifference prices both when public sector's support measures are taken into account (red line) and when they are excluded (blue line).

The blue line shows, for a given price of oil, the maximum value of a corn ton for which ethanol production is profitable. For example, if the crude oil barrel costs USD 70, the generation of ethanol is advantageous if the corn price is below USD 100/ton. In this way, for the combination of prices above the blue line, production is profitable without the intervention of the government.

Table 4

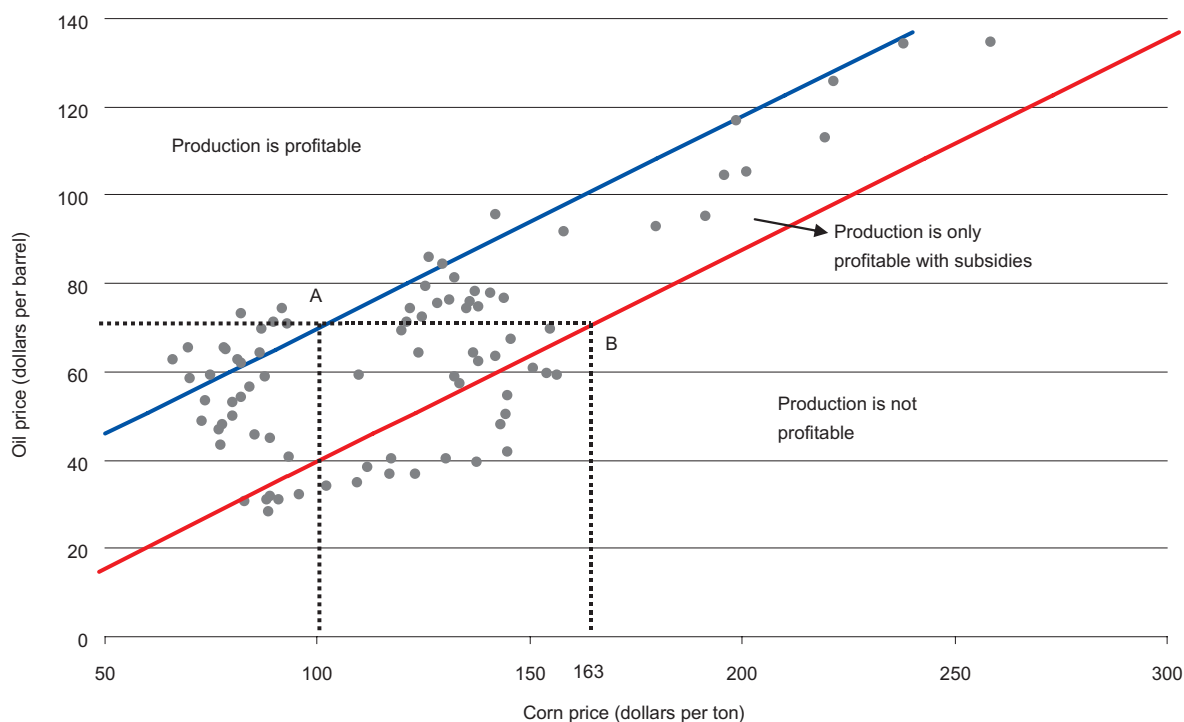
Estimated governmental support to biofuels production and consumption

OECD countries	Estimated total support (billions of dollars)		
	Ehtanol	Biodiesel	Ethanol + Biodiesel
United States	5.4-6.6	0.5-0.6	5.9-7.2
European Union	1.6	3.1	4.2
Canada	0.15	0.013	0.11
Australia	0.035	0.021	0.05
Switzerland	0.001	0.009	0.01
Total	7.2-8.4	3.6-3.7	10.8-12.1

Source: Steenblik (2007).

Figure 26

Oil and corn prices and ethanol production profitability



Source: Author's calculations based on Tyner and Taheripour (2007), FAO (2008) and Bloomberg.

However, when tax incentives are included (which, according to these studies, would be USD 63 per ton of corn), the new indifference prices are represented by the red line. Continuing with the example above, now the production is convenient only when the corn price does not exceed USD 163. Therefore, for the points below this line, biofuel elaboration is not profitable even if supported by the government.

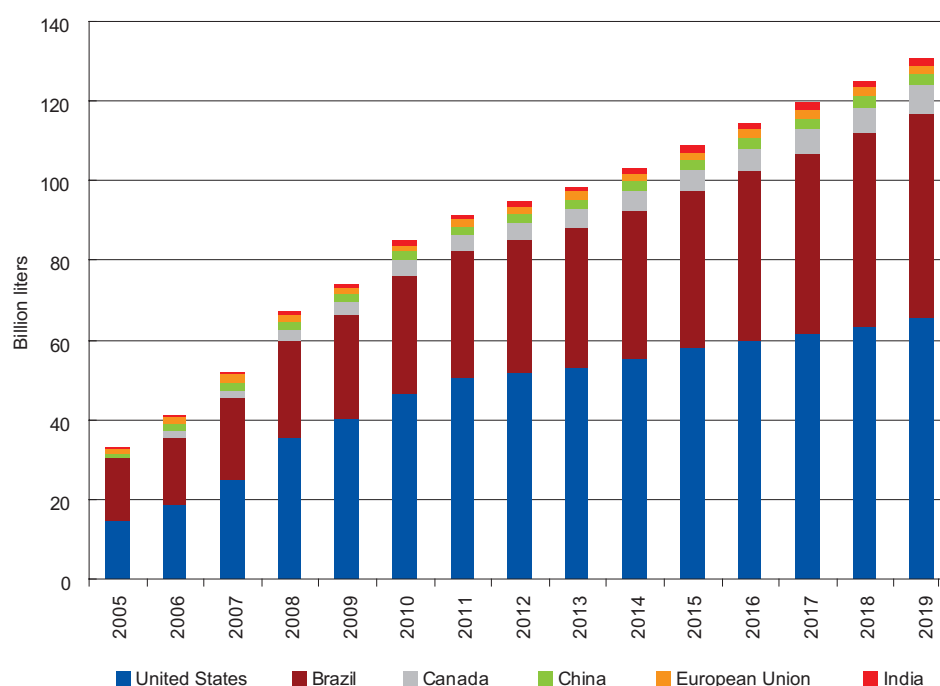
Between both straight lines there is an area where ethanol production is possible only with subsidy support. In addition, Figure 26 shows a scatter plot corresponding to monthly prices effectively recorded between 2003 and 2009. It is possible to see that points are mainly found where generation is profitable only with fiscal incentives or where, even with subsidies, elaboration is not convenient.

These incentive policies have conclusively contributed to the strong growth in biofuel consumption and may indicate that the trend is likely to continue, as shown by the forecasts of Figure 27 and 28 for the leading producing countries. Ethanol volume is expected to increase by 77% in the next decade, while biodiesel would grow by 60% in the same period.

In view of these developments, some analysts have argued that the incentive policies implemented mainly in the United States and the European Union have exerted strong pressure on the agricultural commodity markets, which resulted in an increase in international food prices. The effect of these policies has been modeled in several studies, which have shown sustained price increases in the main commodities for the next decade.²⁵

Figure 27

Forecast of ethanol manufacture by leading producers (2010-2019)

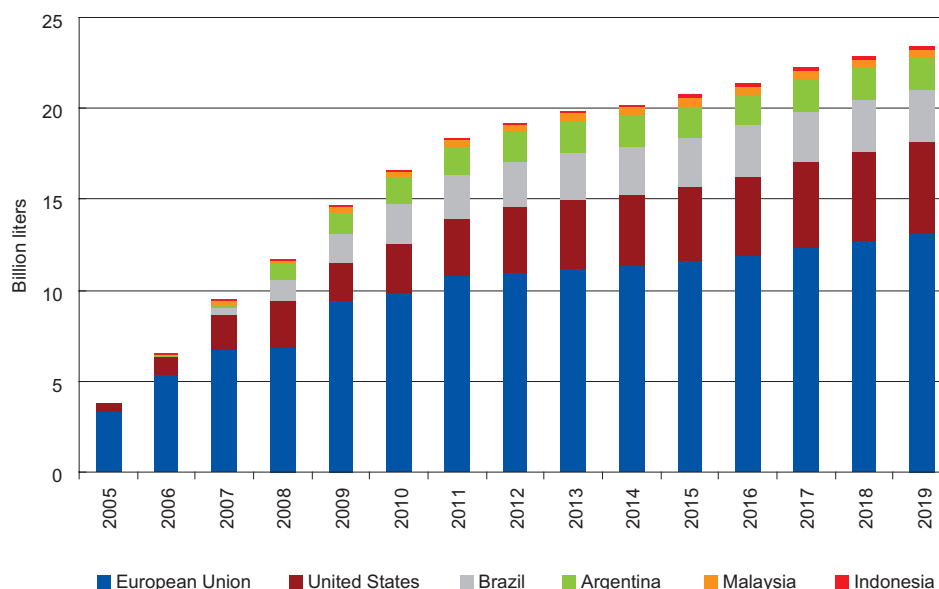


Source: FAPRI (2010).

²⁵ In this respect, see Schmidhuber (2006), Elobied and Hart (2007), Banse and Grethe (2008), among others.

Figure 28

Forecast of biodiesel manufacture by leading producers (2010-2019)



Source: FAPRI (2010).

In this respect, there is an increasing concern about the level of competition generated for resources (mainly land and water) between biofuel and food production and the adverse impact on food supply, market prices and, consequently, on the countries' food security.

Figure 29 illustrates the situation clearly. In the last 15 years, the US percentage of corn production meant for ethanol preparation went up from 4.63% to 35%, while the percentage devoted to human consumption and industrial use went down from 14% to 10%. In absolute terms, the volume processed for biofuels increased 11 times between 1995 and 2009.

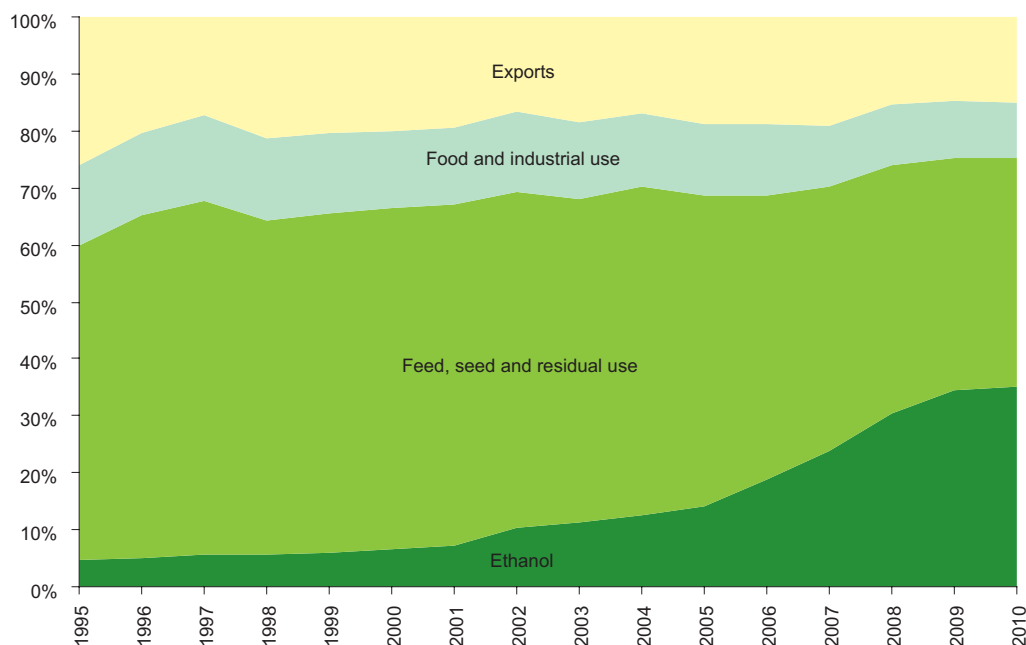
This phenomenon of increasing share of agricultural production oriented to biofuel generation has been verified in other countries as well, such as Brazil with sugarcane and soybean.

Based on this type of evidence, some studies state that the expansion of biofuels, especially in the main food-exporting countries, lies behind global food price increases between 2006 and 2008 and has led to a reduced availability of food supply in international markets (Rosengrat, 2008; Braun, 2008; Msangi et al., 2007).

However, various authors challenge the effects of competition between food and biofuels for the use of resources (Dale, 2007; Kerckow, 2007; Pimentel and Patzek, 2007; Leemans et al., 1996). These papers state that practically all land incorporated to production of crops used in biofuels manufacture has been marginal land and that, in addition, the increase in agricultural productivity has offset, at least partially, the eventual adverse effects of such dispute for resources.

Figure 29

Allocation of corn production in the United States (1995-2010)



Source: Author's calculations based on data from the United States Department of Agriculture.

6.2 China and India

From a historical perspective, the economic transformation process of both China and India is still in its early stages. Nevertheless, this process is sufficiently important to have strong impacts on several aspects of international economy, among them commodity prices may be highlighted, as it will be explained in this section. Both countries have pushed forward radical reforms in their productive structures, with a transition from agricultural economy scarcely integrated to the rest of the world to playing a decisive role in the process of global production and international trade.

To depict this transformation in concrete figures, it is useful to resort to the data from Penn World Table 6.3 (Heston et al., 2009). According to this source, China income per capita relative to United States has four-folded between 1960 and 2007 (from 0.04 in 1960 to 0.16 in 2007), while in India the increase has been around 60% in the same period (from 0.07 in 1960 to 0.11 in 2007). Regardless of the fact that the relative income figure is still far away from that of the world economic leader, two additional facts need to be considered to give a context to the progress made by these countries: (i) the population of China and India has grown more quickly than that of the United States in the same period, as a result of which the total relative income has grown more than the relative income per capita; (ii) the most remarkable progress occurred, in fact, as from the reforms implemented in the late 1970s (China) and early 1980s (India); consequently, their performance is even more outstanding if only the last 30 years are considered.

Regarding this issue, it is worth mentioning that Chinese and Indian performance in the last three decades is similar to other growth “miracles”, as shown by the evidence illustrated in Table 5. There, the economies having grown the most are illustrated per decade (with an analysis of the corresponding percentile in the distribution of world growth), at the same time that there is a control for the persistence of that growth (by

including the condition that in most years of each decade, the growth of these countries has exceeded the world average). This remarkable performance of China and India may be confirmed in Figure 30, where it is compared to that of other economic blocks.

Table 5

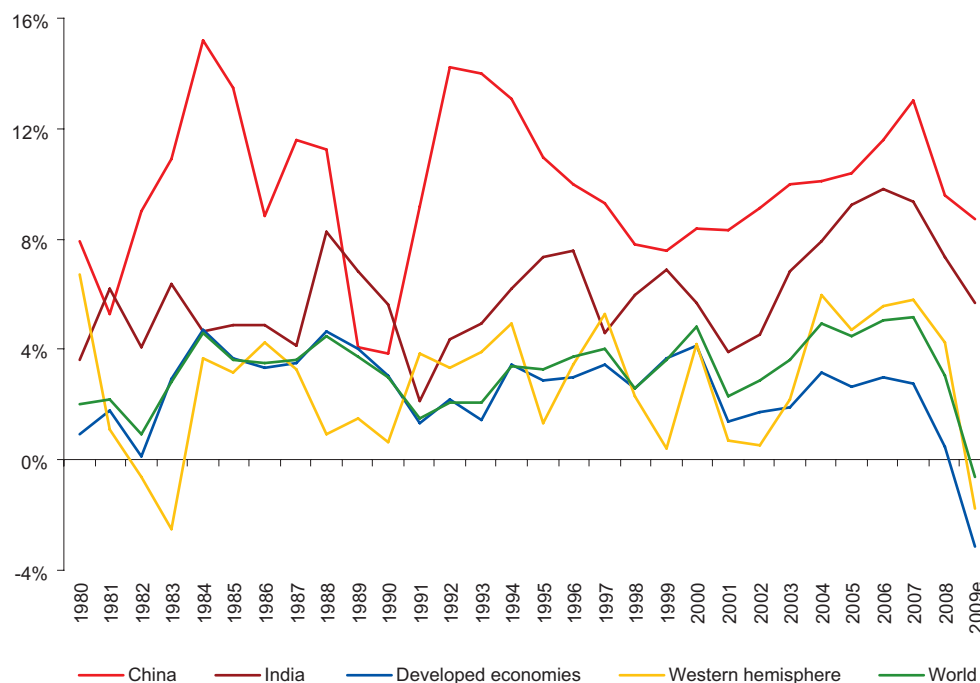
Experiences of high and sustained growth for decades, 1960-2007

Period	Growth Pattern	Accumulated growth in the period above 90 percentile of distribution	Accumulated growth in the period above 75 percentile of distribution
1961-1970	10 - 9 years of growth	Japan, Greece, Romania, Iran, Spain, Gabon, Hong Kong	Mauritania, Morocco
	8 – 7 years of growth	Italy, Austria, Portugal, Puerto Rico, Thailand, Turkey, France, Papua New Guinea, Cyprus, Botswana, Togo.	South Korea, Israel
1971-1980	10 - 9 years of growth	Botswana, Romania, Ecuador, Indonesia, Taiwan, Malaysia	Syria, Hong Kong, Cameroon
	8 – 7 years of growth	Brazil, Paraguay, Singapore, Guatemala	Panama, South Korea, Thailand, Greece, Gabon, Ireland, Lesotho
1981-1990	10 - 9 years of growth	South Korea, Taiwan, Hong Kong, China , Botswana, Thailand, Cyprus	Egypt, Singapore
	8 – 7 years of growth	India , Pakistan, Japan	Turkey, Malaysia, Nepal, Indonesia, Finland, England, Sri Lanka, Puerto Rico, Ireland, Portugal
1991-2000	10 - 9 years of growth	China , Taiwan, Mauricio, Singapore, Chile, Malaysia, Uganda, Sri Lanka	South Korea, Ireland
	8 – 7 years of growth	Denmark, India , Guatemala, Australia, England, Guinea-Bissau, Portugal	Thailand, Cyprus, Trinidad and Tobago, Dominican Republic, Puerto Rico, Mali, Lesotho
2001-2007	7 years of growth	Mozambique, Romania, China , Trinidad and Tobago, India , Tanzania, Lesotho	Nigeria, Zambia, Chad
	6 – 5 years of growth	Thailand, Greece, Honduras, Bangladesh, Finland, South Africa	Botswana, Mauricio, Sri Lanka, Turkey, Malaysia, Hong Kong, Congo Democratic Republic, Malawi

Source: Author's calculations based on PWT 6.3.

Figure 30

Annual growth rates of selected countries (1980-2009)



Source: Author's calculations based on IMF data.

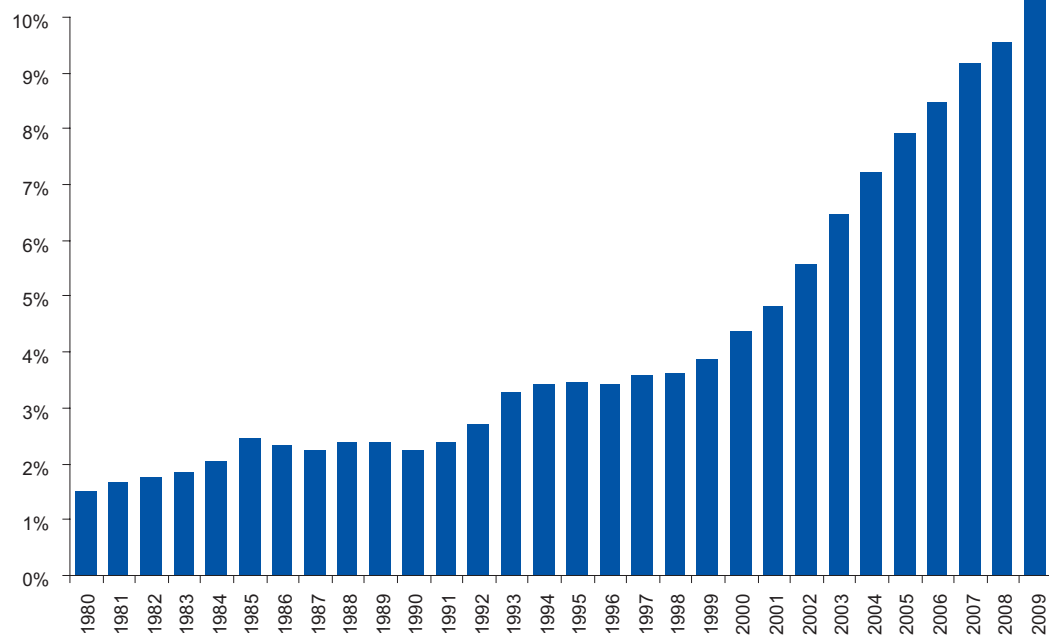
According to IMF data, as a corollary of this systematic and sustained growth, China and India account for about 17% of world GDP measured in PPP dollars (10% if measured in current dollars) and are the main consumers of several commodities.

In turn, as a result of the gradual openness generated by the reforms implemented in the early 1980s, the share of both economies in international trade went up from being slightly over 1% in 1980 to over 10% in 2009 (Figure 31).

Clearly, the emergence of two new and influential players for global trade implies new challenges in the composition of world trade flows. One of the most remarkable is the role they are playing as importers of numerous commodities. To mention just a few examples, Figures 32 to 34 illustrate the evolution of the Chinese and Indian share in world imports of soybean, iron ore and oil.

Figure 31

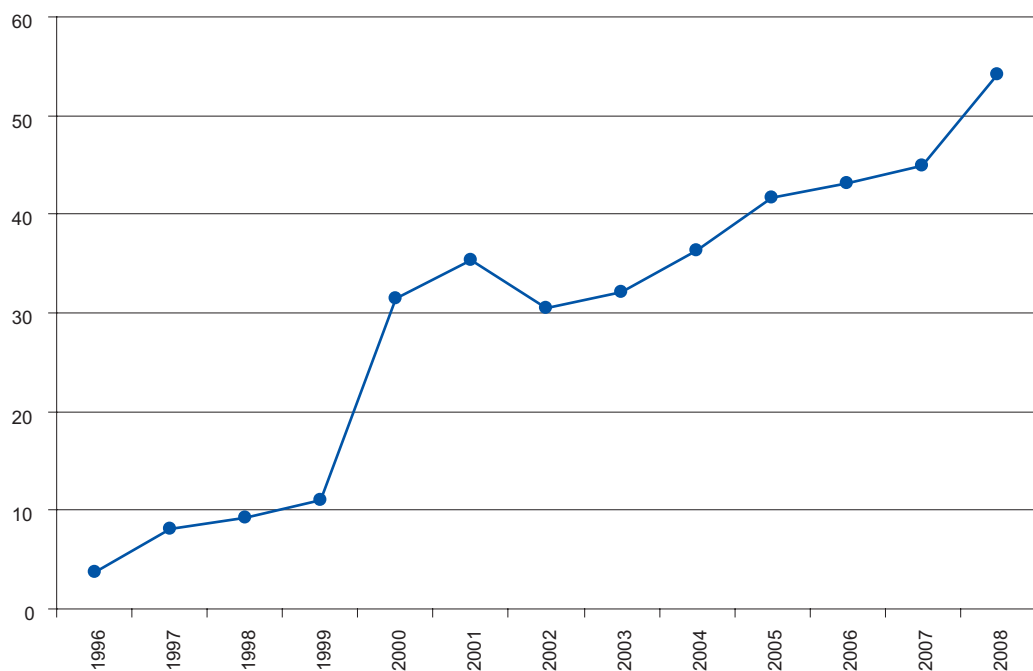
China's and India's share in international trade



Source: Author's calculations based on IFM data.

Figure 32

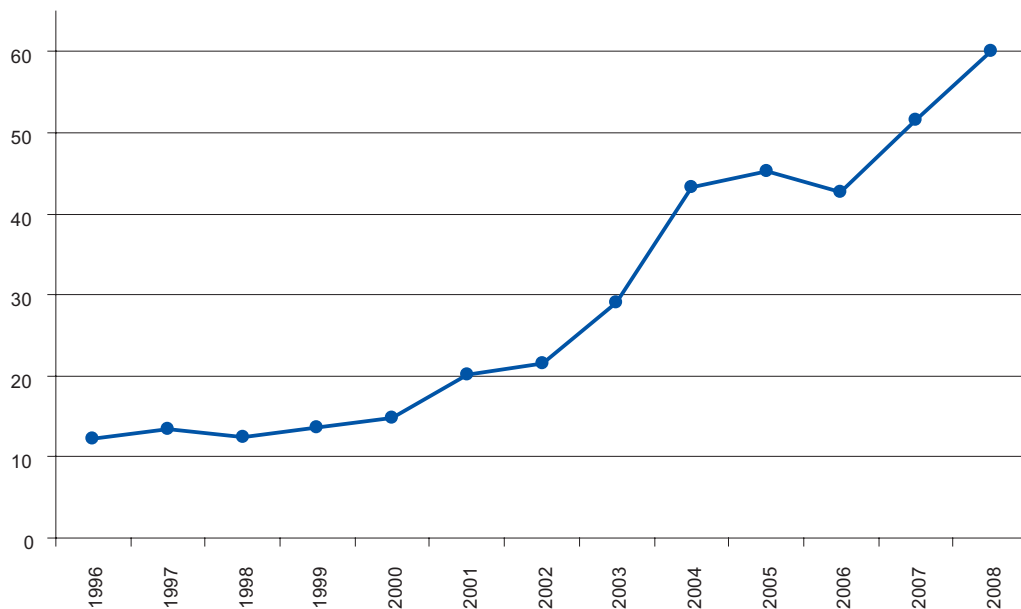
Soybean imports by China plus India (soybean world imports %)



Source: Author's calculations based on Comtrade data.

Figure 33

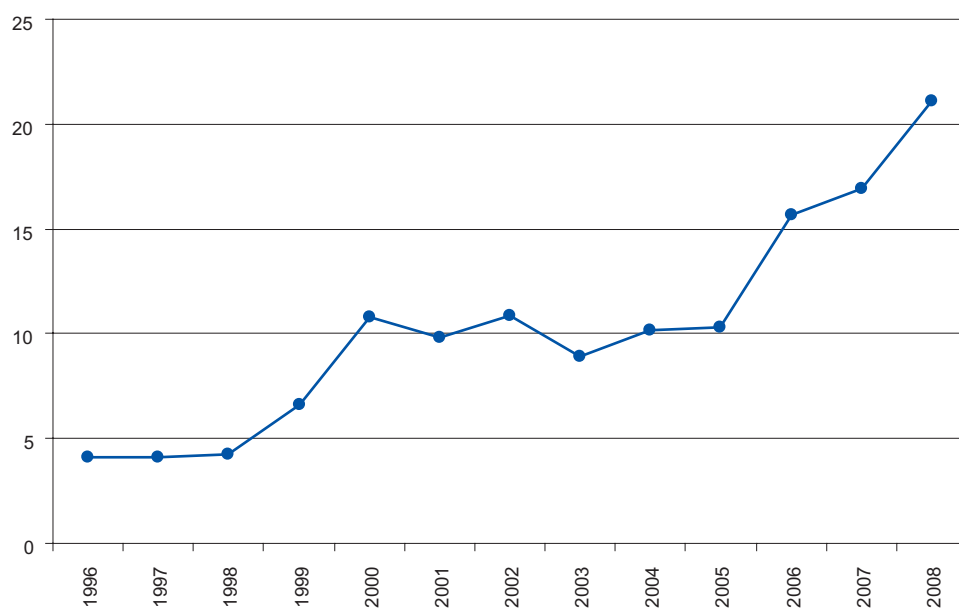
Iron imports by China plus India (iron world imports %)



Source: Author's calculations based on Comtrade data.

Figure 34

Crude oil imports by China plus India (crude oil world imports %)



Source: Author's calculations based on Comtrade data.

Thus, the incorporation of China and India to world trade means that an important number of commodities (especially metals, energy-related products and soft commodities such as soybean) are facing a sort of “minimum demand level”, at least in the short and medium term, which will probably lead to higher prices than those recorded during the past decade.

In the next section, an analysis of the incidence of China and India on food-related commodities and the situation for metals is presented.

6.2.1. China and India: Growth and nutritional transition

According to studies by Popkin (1993) and Popkin et al. (2001), the expression “nutritional transition” makes reference to the changes in food consumption patterns as the income per capita increases.

According to these authors, in the first stage of this process, the improvement of income per capita from very low levels is associated with a sharp increase in food consumption by each inhabitant. In the second stage, and as the income per capita continues to grow, calorie intake goes up and consumption composition diversifies: grain-based food loses ground in relative terms against the consumption of meat, fish, fruit and vegetables.

These changes were originally experienced by several countries which are now considered as developed and took place during a long period of time, encompassing several decades. However, in economies like China or India, this phenomenon is occurring at a clearly more accelerated pace.

In addition to the shifts in consumption patterns caused by changes in consumers’ income, the effects of a greater urbanization must be underlined, since it has an especially strong impact on the level and composition of food consumption in a specific country if the income of urban consumers is considerably higher than that of rural workers. Against this backdrop, rural workers will increase their calorie intake but will preserve their traditional consumption habits, while city workers will reach a calorie intake level closer to that of developed countries because their eating habits tend to change.

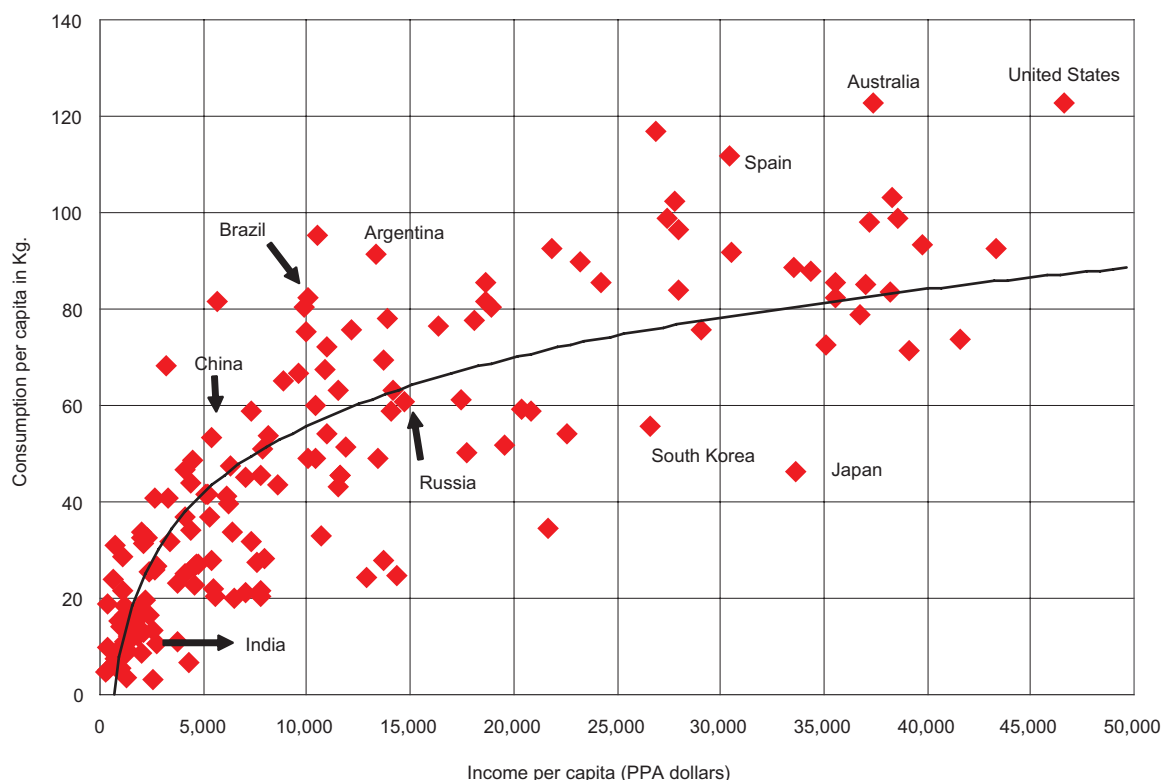
Moreover, a higher degree of urbanization means a higher share of labor in GDP, which also impacts consumption patterns. According to Schmidhuber and Shetty (2005) and King et al. (2000), people living in the cities have a higher number of meals outside their home; consequently, when the urban population increases, the consumption of food of animal origin also tends to augment (Rae, 1998). As an example, it is worth mentioning that, according to the forecasts of meat demand by Chinese consumers, China will account for an increase of meat world demand of around 40% between 1997 and 2020 (Rosengrant et al., 2001).

As a correlation of these trends, the strong rise in meat consumption will result in a higher demand of soybean, due to its increasing use to feed the animals. Likewise, the remarkable increase in the per-capita consumption of vegetable oils in urban centers, mainly soybean oil and palm seed oil, will also unleash an additional pressure on the oilseeds market.

In this sense, the paper by Delgado et al. (1999) asserts that as an economy develops, spending on meat consumption also increases significantly (accounting for a strong elasticity). Figure 35 illustrates that this positive relation between income and consumption is not linear because elasticity depends on the income

level: when the income per capita increases, at low income levels meat consumption raise more than proportionally up to the point in which income reaches approximately USD 4,000-5,000-PPP, from there on, it increases less than proportionally.

Figure 35
Meat consumption and income per capita (2007)



Source: Author's calculations based on FAO and IMF data.

Beyond these general features, the changes in food consumption patterns occurred both in China and India in the last decades have shown differences. This is illustrated in Table 6, which documents the modifications in the composition of the food basket (per main items) in both countries between 1990 and 2005.

Due to a cultural tradition that favors the consumption of vegetables, India's meat intake has remained virtually constant, while China's has grown over 25%.

Cereals consumption marks another difference between both countries. It decreased slightly in India, with a moderate rise in fruit and vegetable consumption. Instead, in China, these changes were remarkable: cereals intake went down sharply to a half, from 42.7% in 1990-1992 to 21.2% in 2003-2005, while the consumption of vegetables went up from 21.5% to 35.9% in the same period.

These data would be indicating that protein intake has changed moderately in India and considerably in China. This means that, though the high population growth and the significant increase in the income per capita have resulted in higher absolute levels of food consumption in India, the country is still in the first stage of the nutritional transition; therefore, it has a great potential for an accelerated growth in calorie intake.

Table 6

Composition of the food basket of China and India (% of total, 1990-2005)

Product	China			India		
	1990-92	1995-97	2003-05	1990-92	1995-97	2003-05
Alcoholic Beverages	3.0	3.8	4.0	0.3	0.4	0.4
Cereals	42.7	32.8	21.2	42.6	40.2	36.6
Eggs	1.5	2.3	2.6	0.3	0.4	0.4
Fish and seafood	2.5	3.7	3.4	1.1	1.1	1.2
Fruits	3.9	5.9	7.2	7.4	8.5	7.9
Meat	5.9	6.8	7.6	1.2	1.2	1.3
Milk	1.3	1.3	2.7	13.9	14.8	15.5
Oilcrops	1.1	1.2	0.9	1.9	1.9	1.7
Pulses	0.4	0.2	0.1	3.4	3.2	2.8
Starchy Roots	12.1	10.5	9.8	5.2	5.5	5.2
Sugar and Sweeteners	1.7	1.4	1.0	5.9	5.7	5.8
Vegetable Oils	1.2	1.2	1.4	1.7	1.9	2.4
Vegetables	21.5	27.2	35.9	13.8	13.7	17.2
Others	1.3	1.6	2.2	1.4	1.5	1.7
Total	100.0	100.0	100.0	100.0	100.0	100.0

Source: Author's calculations based on FAO data.

China, in turn, has modified its food consumption pattern at an accelerated pace and, in the last two decades, it has come closer to the calorie levels of developed countries. Therefore, since China is already in the second stage of the nutritional transition, it is likely that future shifts in the consumption basket will not be as impressive.

In addition, it is worth mentioning that the demographic factors for the next years will tend to deepen India's increasing relative weight, especially because the birth control policies adopted in China will result in a stagnated growth of the work force. The opposite is likely to occur in India, where labor supply is expected to increase by 135 million people in the next 10 years. Thus, these demographic trends, added to the accelerated improvement in the income per capita and its lagged nutritional transition, are positioning India as the economy with the best potential to create "new demand" of food in the next decade and, consequently, to play a key role in the soft commodity market.

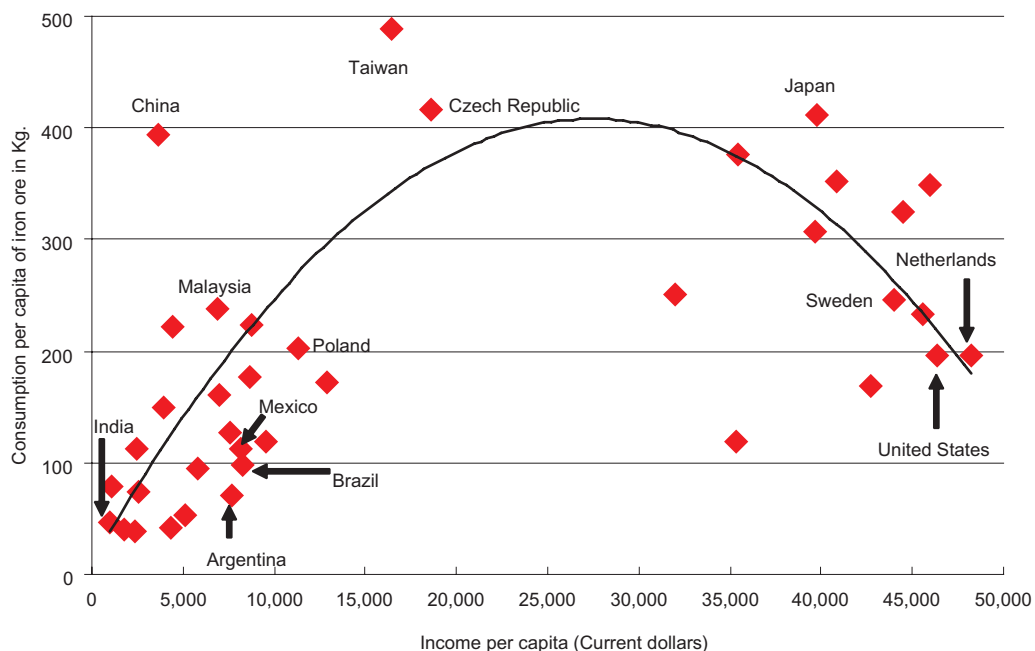
6.2.2 China and India: Intensity of use of metals and energy

The use of metals has also grown steadily in recent decades both in China and India, as evidenced by their relevance as importers of several metals in the international markets (such as copper and steel).

As pointed out in section 2.2., several authors have raised the hypothesis of the "intensity of use" (Malembaum, 1973; Syrquin and Chenery, 1989), which establishes an inverted-U ratio between metal use and income level. Figures 36 and 37 confirm the evidence found by the above-stated IMF study (2006): the metal intensity of use increases to some extent as the GDP per capita grows and then it starts to decline.

Figure 36

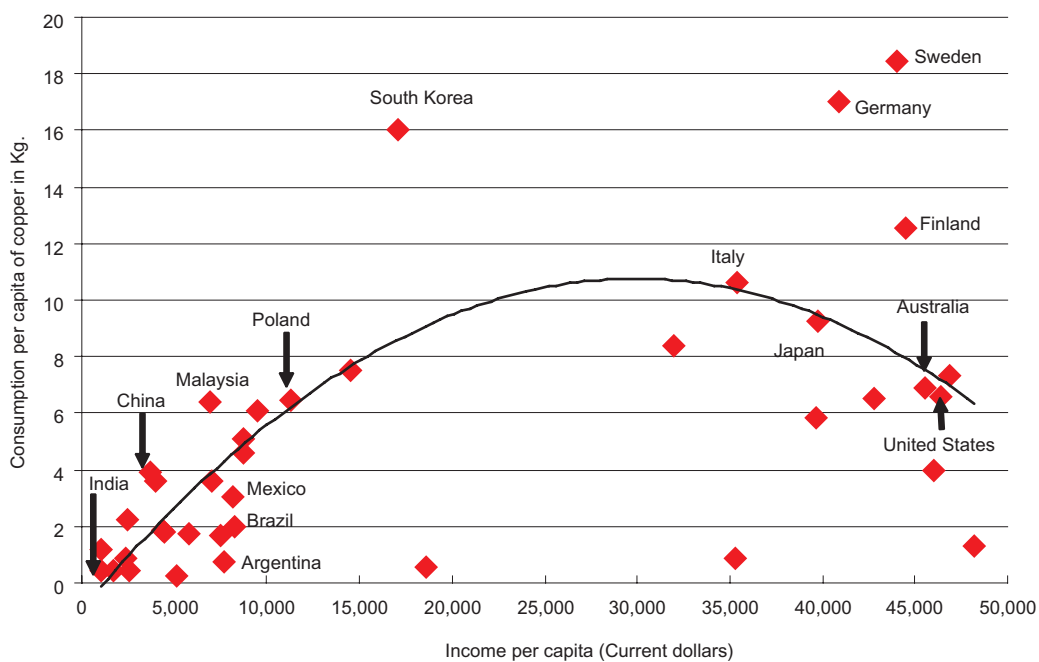
Iron ore consumption and income per capita (2009)



Source: Author's calculations based on data from the IMF and the World Steel Association.

Figure 37

Copper consumption and income per capita (2008)



Source: Author's calculations based on data from the IMF and the Copper Study Group.

Thus, the poorest countries, or rather those in an early industrialization stage, have a very low use intensity of metals since these economies have poorly mechanized production systems or simply live on subsistence farming.

As countries reach higher levels of development, a substantial increase in infrastructure investment (both public and private) is required, and this is related in general to the use of metals. Likewise, there is a second factor associated with a better standard of living, which in turn increases the demand of metals: a higher consumption of durable goods.

However, this higher use of metals required by the transformation process of the productive structure (in the transition towards industrialization) reaches a peak as from which it starts to decline, a phenomenon which may be due in part to the efficiency improvements in the use of inputs.

Consequently, if we extrapolate the past performance of China and India as regards growth, and if the ratio of Figures 36 and 37 is confirmed, a remarkable increase in metal per-capita consumption is likely to occur, which would contribute to keeping their prices high in the medium and long term.

As a general conclusion of this section, it is worth highlighting that evidence proves the importance of both biofuel production and the structural changes experienced by China and India for commodity prices, especially in the last 10 years. In this sense, it is necessary to remark that even though standard variables from literature are included to measure the demand of commodities, it might be considered that the factors described in this section are only partially captured in the empirical model proposed. Consequently, it is advisable to be cautious when designing future scenarios for commodities which are exclusively based on our empirical estimates.

7. Conclusions and Policy Implications

From the policymaker perspective, the distinction between permanent and temporary movements of macroeconomic variables is one of the greatest challenges for appropriate decision-making. One of the main objectives of economic models and empirical estimates is to provide a consistent framework to make these decisions reasonable enough. To this effect, the proposal of this paper is to use a nonlinear multivariate methodology for a better understanding of the causes behind commodity price movements, once an explicit role for the financialization of the commodity markets is considered.

It is interesting to think about financialization as an amplifying factor of commodity price cycles. For this purpose, a conceptual framework is developed where fundamentals and financialization interact, dealing with the speculative activity as an element that mainly affects short-term price dynamics rather than long-term equilibrium.

Therefore, in the model proposed, fundamentals continue to be the only real force to account for long-term price behavior. However, the heterogeneous expectations of commodity market participants are crucial for the definition of the adjustment-to-equilibrium characteristics.

Regarding the fundamentals, the relevance of key macroeconomic variables for the determination of the equilibrium price is confirmed, namely the United States real exchange rate, the aggregate industrial production as proxy of world demand and the international real interest rate. There is also a negative conditional correlation between the commodity price index considered and the real return of stock markets.

Short-term dynamics exhibit the most outstanding results. The evidence found supports the idea that commodity prices tend to correct towards equilibrium, but this correction only occurs if the past misalignment is high enough. Consequently, under a low misalignment regime, the corrective forces do not prevail and prices may move in any direction, according to market sentiments.

Likewise, the theoretical and econometric analysis in this document is supplemented by a qualitative study of two recent structural changes which are undoubtedly the most relevant for the commodity markets: the development of biofuels as supplementary source to fossil fuels and the accelerated rise of China and India as new economic powers. In both cases, these factors cannot be ignored when considering perspectives and designing scenarios for commodity prices, at the same time that they are difficult to be fully captured in an empirical model as the one proposed.

As regards policy implications, this paper suggests that developing countries that depend heavily on commodities should pay proper attention to the misalignment of their prices: reversals tend to be sharp when the gap between the observed price and the fundamental price exceeds approximately 25%. It should be kept in mind that commodities affect key economic variables in this type of countries, such as GDP growth, inflation, income distribution, national savings or tax revenue.

It is also important to point out that the evidence gathered proves that the factors affecting commodity prices (such as the real international interest rate and the US real exchange rate) are similar to those affecting capital flows. This explains why it is difficult for developing countries to lessen terms of trade shock effects with external financing. The same fundamentals that worsen the terms of trade have an adverse impact on the access to international credit markets. In this case, an appropriate combination of policies should include structural measures tending to smooth the external cycles and reduce the dependence on commodities when prices reach high levels.

Appendix. Impulse-response analyses

Impulse-response analysis considering three-standard deviation shocks.

Figure 1A

US Effective real exchange rate shock

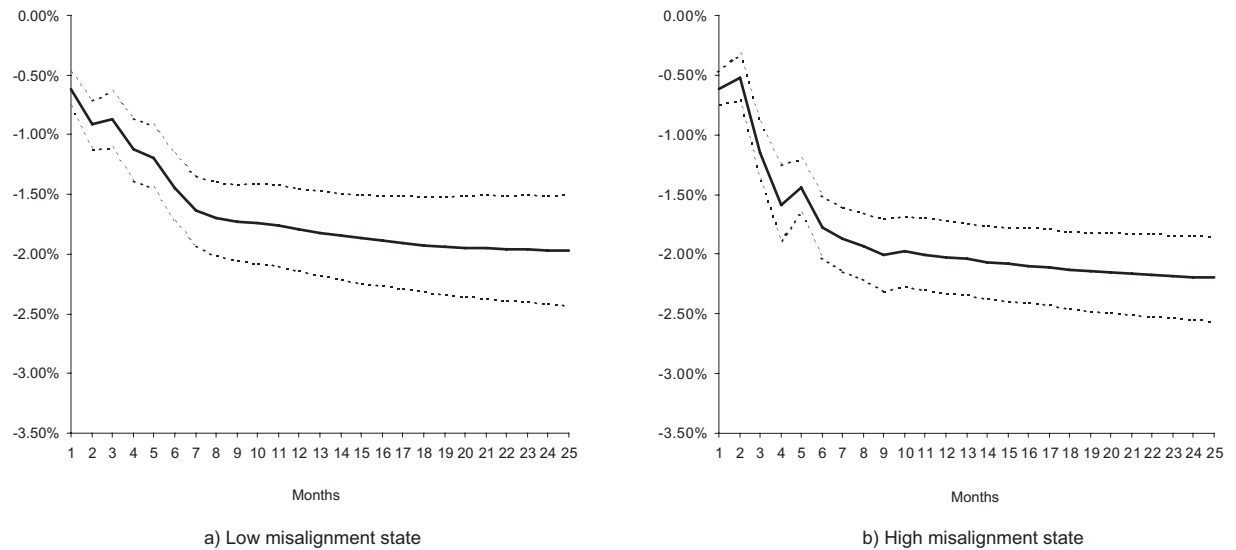


Figure 2A

Industrial production shock

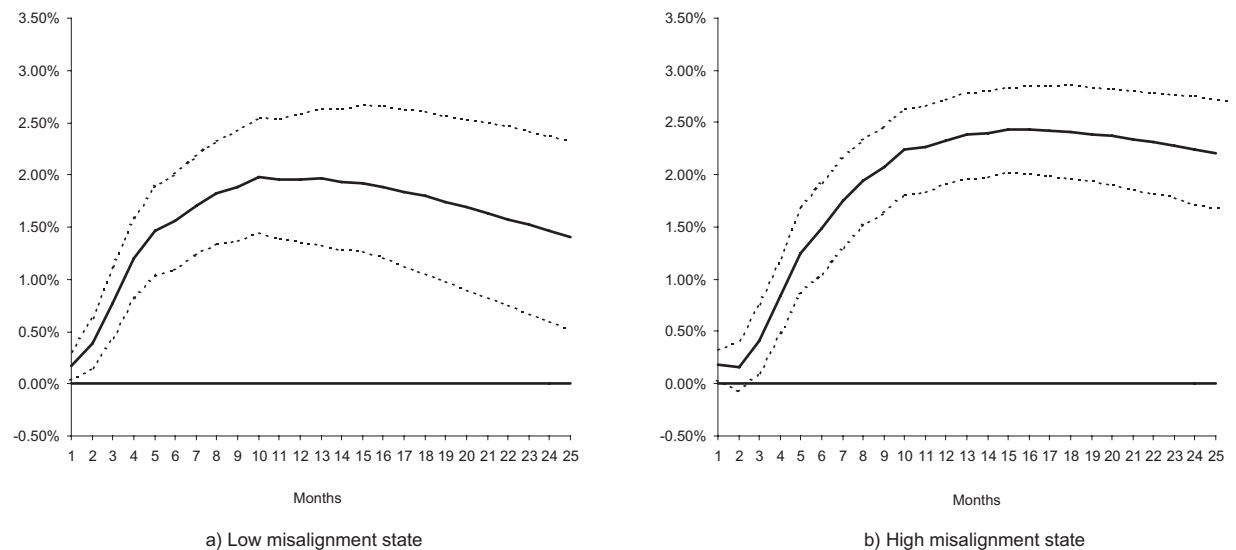
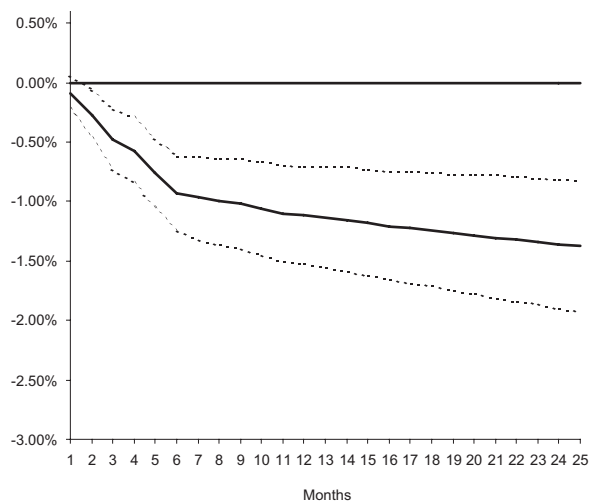
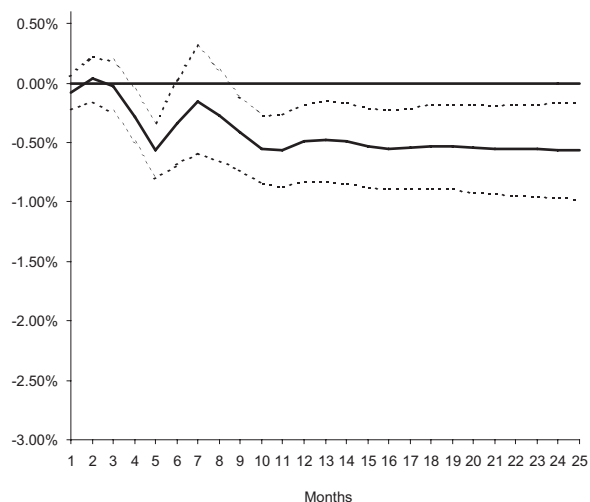


Figure 3A
US real interest rate shock

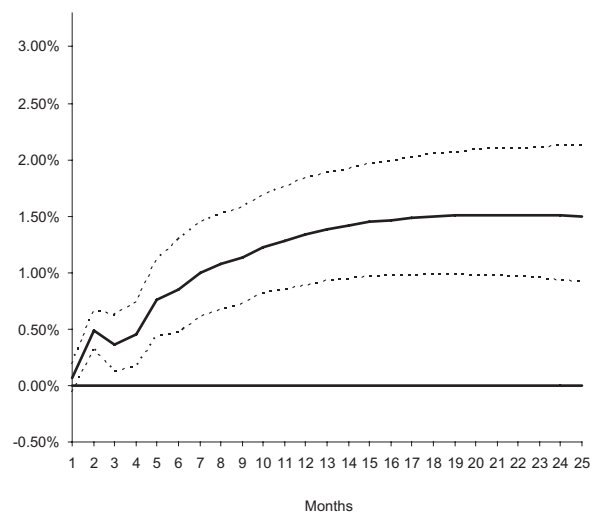


a) Low misalignment state

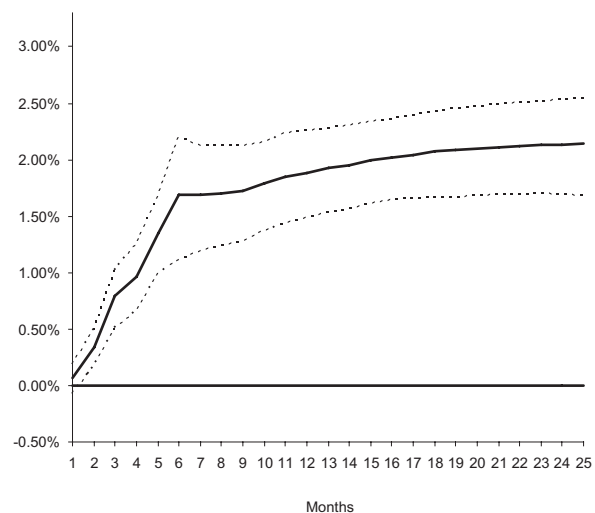


b) High misalignment state

Figure 4A
Real Dow Jones index shock



a) Low misalignment state



b) High misalignment state

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The issue of commodity prices has regained prominence as a consequence of the abrupt swings in commodity prices in recent years. The present study contributes to the debate using a theoretical and empirical model that captures both short-run and long-run factors affecting commodity prices. The so-called process of financialization of commodity markets is also analyzed, suggesting that this element could generate a non-linear adjustment pattern of prices towards their equilibrium values. In order to support this hypothesis, a heterogeneous agent model is developed. The empirical evaluation is carried out by means of a smooth transition vector autoregressive model. The results suggest that high misalignments between spot and equilibrium prices tend to correct relatively fast, whereas small gaps tend to persist over time without any correcting force taking place. Finally, the paper analyzes the main consequences for commodity prices of two ongoing structural transformations, namely the production of biodiesel and the rapid growth path followed by India and China.

